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Something else Warsh said at Jackson Hole

August 31, 2026

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Executive summary

Kevin Warsh used his first Jackson Hole address as Chairman to make an inflation-forward case, and the market heard it. Yields on two-year treasuries jumped 11 basis points on the day and the implied probability of a September interest rate hike jumped from 35% to 58%.

The longer-term implications for sovereign debt were less talked about. This piece looks at some of the deeper issues behind the megatrend of sovereign debt – particularly given Wash's interesting comment that "Trends matter most". Indeed, our megatrend framework identifies sovereign debt as the most concerning negative trend facing developed economies over the medium term.

Yet, our proprietary data shows that people are more interested in the equity market than bond market. Unusually, that preference extends even to periods of economic uncertainty. Indeed, 28% of Americans would buy equities during periods of economic uncertainty while only 24% would buy bonds. In the UK, the same picture exists with figures of 25% and 18% respectively.

We have previously shown that since interest rates normalised in 2022, the number of Vix spike events per month has risen 51% versus the pre-covid level. More shocks are arriving into a market that is less willing to own the traditional shock absorber.

The question is whether the potential productivity gains from AI will substantially alleviate the negative weight of the sovereign debt burden in many developed countries. Our tech megatrend indicates this is possible, and so does history. However, for the optimistic scenario to eventuate, the positive impacts of AI on the economy need to accelerate further. That leaves the medium-term distribution of outcomes for sovereign debt wide and unusually asymmetric.

Something else Warsh said ...

The market reaction last Friday was focussed on the here and now. Warsh argued that "the responsibility for 65 months of sustained, elevated inflation sits squarely with the central bank", and remarked that "we have work to do" whilst pointing to PCE inflation running at 3.7% over twelve months and 4.1% over six. Markets moved the September probability materially higher and two-year yields jumped 11bps.

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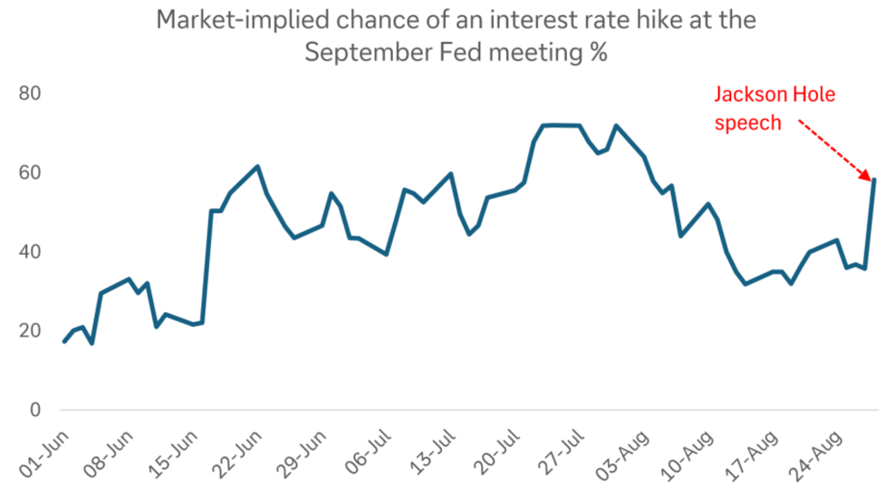
Figure 1: An inflation-forward speech spiked the chance of a September rate hike



Source: Bloomberg Finance L.P., Deutsche Bank Research

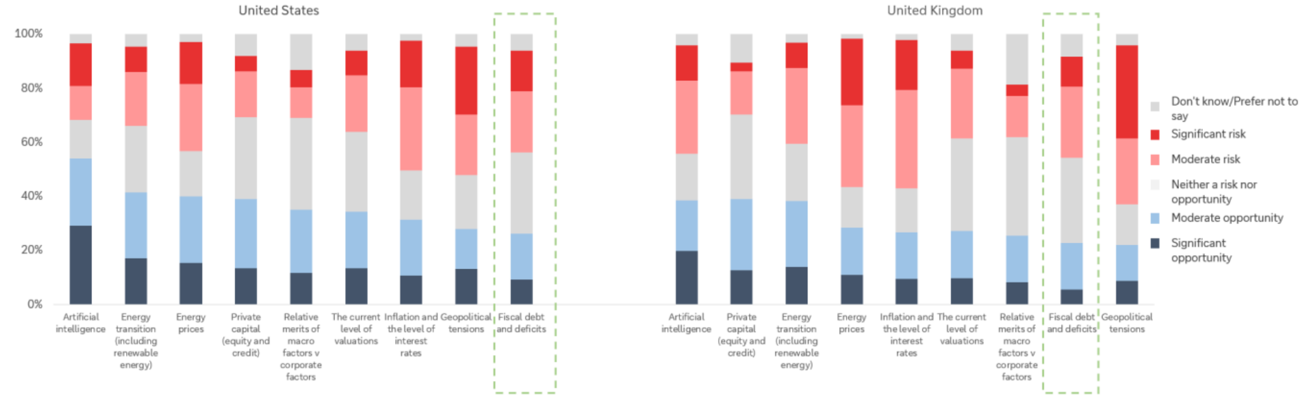
Something else Warsh said received very little coverage – “Trends matter most”. It was easy to interpret this as referring to the current trajectory of short-term inflation. However, it is possible, and even likely, that he was referring to the Fed’s decision making around sovereign debt as a whole – across the short, medium and long term.

Indeed, the current situation of US sovereign debt and its trend is one of the key reasons why Treasury Secretary Bessent has previously said he wants to target the 10-year yield. The current buyback programme falls into line here. The key thing here is the credibility of the fiscal trajectory because, as future shocks arise, the way the bond market performs will affect broader portfolios.

Our worry is how investors view bonds

A proprietary survey from our dbDataInsights team found that the average person in the US and UK sees more opportunity in equities rather than bonds. First, when investors assess fiscal debt against other assets and themes, they see a lot of risk and fewer opportunities. The sell-off in bonds over the last 4 years (following a 40-year bull market) has certainly contributed.

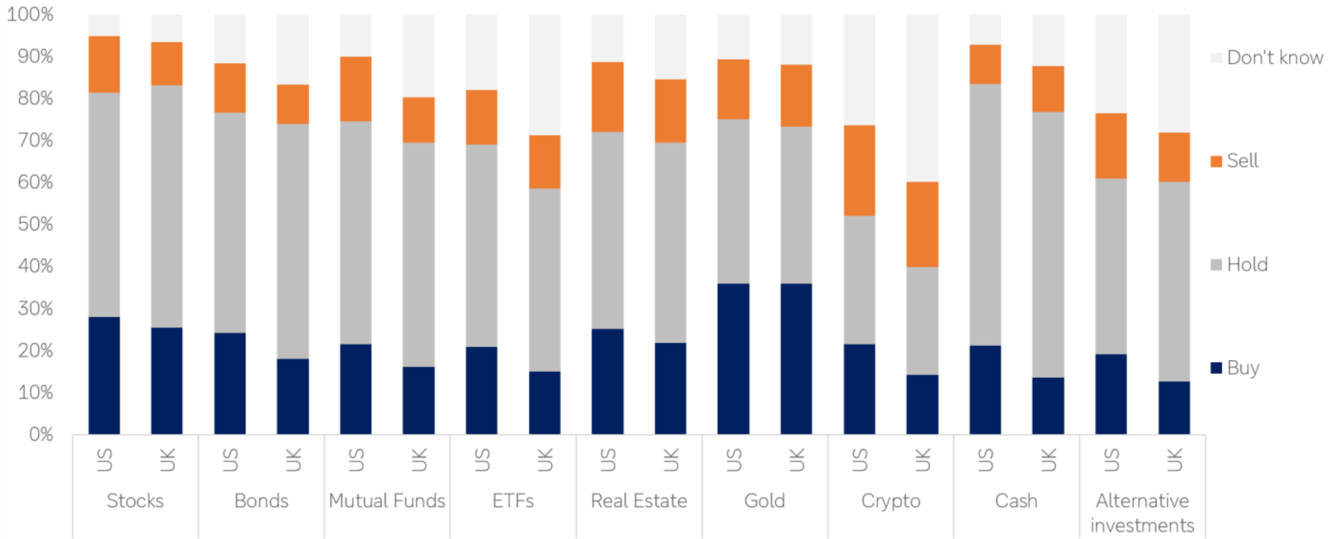
Figure 2: Survey question: “For each investment theme listed below, please indicate whether you perceive it as a risk or an opportunity for investors in 2026"



Source: dbDataInsights

Second, in times of economic uncertainty, investors are more likely to want to buy stocks than they are to buy bonds. This is a reversal of the textbook relationship between the two:

Figure 3: Survey question: 'In any period of economic uncertainty, what do you think is the best strategy for each of the following investment asset categories?”



Source: dbDataInsights

This tilt away from bonds is not necessarily irrational. Since 2022, investors have found that bonds can lose significant value at the same time that equities fall. Our own work on haven assets shows that, since 2020, the combination of six traditional haven assets (gold, dollar, Swiss franc, yen, 10yr treasuries and bunds) has failed more often than it has worked, and it failed specifically during the four big shocks of the decade (Covid, rate rises, tariffs, Iran).

The worry into the medium term is whether investors continue to see the bond market this way as sovereign debt positions across most developed countries continue to worsen. That will affect investor portfolios given that bonds are supposed to be a counterpoint that rallies when everything else falls and avoids investors being forced to sell risk assets into weakness.

Why a less-loved bond market is a problem: uncertainty is rising

We have shown previously that since interest rates normalised in 2022, the number of VIX spike events per month has increased by 51% compared with the pre-Covid level. This follows from our broader megatrend framework that tells us that as interest rates have normalised in the 2020s, megatrends have re-emerged as a defining force in economies and markets for the first time since the 2008 financial crisis.

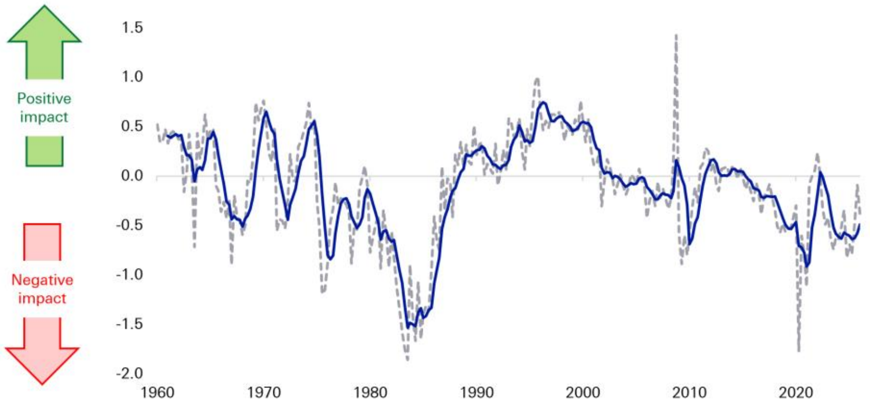
So, we have more shocks arriving, a haven complex that has been unreliable through the shocks we have already had, and an investor base that says it would rather buy equities than bonds when things become uncertain. That is a fragile configuration. It is also one in which the sovereign debt megatrend can do damage. That is, not through a gradual repricing, but through unpredictable sovereign debt sell-offs, particularly triggered by exogenous shocks and investor emotion.

Sovereign debt is the most concerning megatrend in our framework

Our megatrend model tracks six global forces: technology, sovereign debt and deficits, geopolitics and globalisation, domestic politics and social discontent, demography, and energy disruption and transition. We use almost 100 data series aggregated into a quarterly indicator for each megatrend, all standardised so they can be compared and combined.

Of the six, sovereign debt is the one we are most concerned about. Indeed, our sovereign deficits indicator has been trending down for three decades. The deterioration is being compounded by demography: working-age populations are shrinking across most of the G20, which simultaneously weakens the revenue base and strengthens the claim on the spending side.

Figure 4: Our sovereign debt megatrend indicator and its impact on economies and markets



Source: Bloomberg Finance L.P., Haver Analytics, Factset, Deutsche Bank Research

In terms of where debt is heading, in the US, the CBO has public debt rising from around 100% of GDP to 120% by 2036, with deficits pushing over 6% of GDP. Most other developed economies face a version of the same problem. Aside from declining demographics, the deterioration in sovereign debt in interconnected with domestic politics and social discontent, and geopolitics.

Even our optimistic scenario only reaches neutral

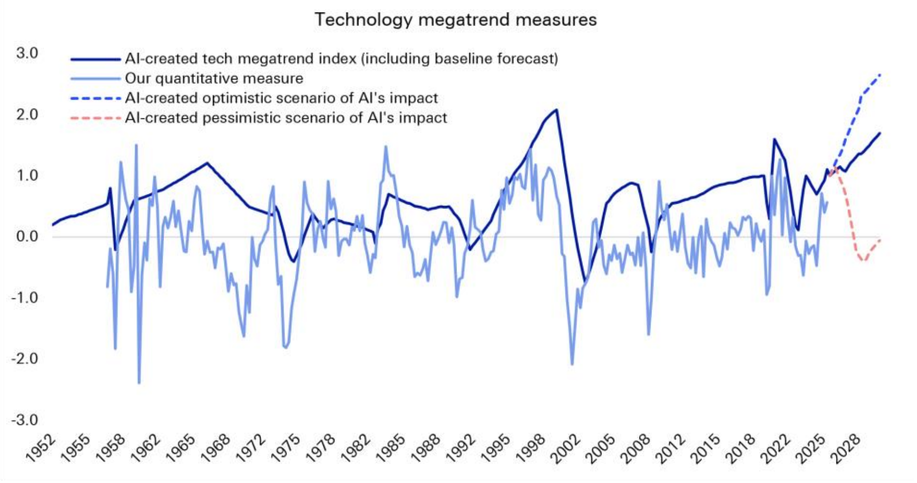
We ran the sovereign debt megatrend forward to 2030 under baseline, optimistic and pessimistic scenarios. In the optimistic case — where AI adoption accelerates, productivity growth lifts materially, government spending contributes positively to GDP, and global demand for treasuries holds up — the indicator moves from having a clearly negative impact on markets and economies to having a broadly neutral one.

None of our scenarios move the sovereign debt megatrend materially into positive territory. That asymmetry should shape how the medium-term risk is positioned. The best realistic case for sovereign debt over the next five years is that it stops being a drag.

Can technology rescue us? Warsh asked the right questions

Technology is the one megatrend currently making a clearly positive contribution to global economies and markets. And our analysis points to AI being significantly more impactful than the internet of the 1990s. Indeed, whenever our technology indicator has accelerated to a peak, productivity growth has accelerated sharply behind it: to 3-4% for sustained periods in the 1990s. Our baseline has the indicator exceeding its 1990s peak by 2030, which implies productivity growth potentially above that range.

Figure 5: Should our tech indicator continue to climb as we expect, the implied productivity boost will go some way towards offsetting the worsening sovereign debt position.



Source: Bloomberg Finance L.P., Haver Analytics, Factset, Deutsche Bank Research

That is the escape route from the debt overhang and Chair Warsh's remarks included several references to this. He asked, "Will the application of AI cause a significant, sustained rise in productivity across the economy? He also pointed to the conversation between AI being "complementary or competitive to labour" He also asked, "Will the next generation of AI models demand even greater capital intensity, or will the models themselves help devise a capital-light solution?"

We would draw attention to the capital-intensity question in particular, because it has a bearing on sovereign debt. In the future, a capital-hungry AI build-out may compete with the government for savings at exactly the moment when the government's borrowing requirement is rising. The same technology can therefore either compound the sovereign debt problem in the near term or relieve it, depending on which way the question resolves.

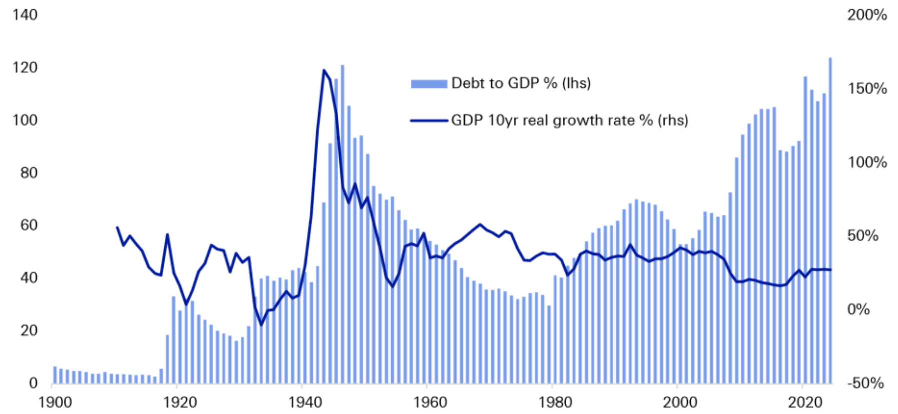
Warsh is waiting for his task forces before committing to answers, and has said explicitly that their recommendations "will come later and have no bearing on decisions we make in the current policy conjuncture".

History says a bad sovereign debt position can be overcome

None of this is a prediction of crisis, and history is on the side of the optimists more often than the mood suggests.

The clearest parallel is the "Long Depression" in the US in the mid-1870s, which followed a financial crisis and investor fear at the overbuilding of railroads — a capital-intensive general-purpose technology financed enthusiastically and then abruptly distrusted. There was little government intervention and unemployment likely topped double digits. Yet towards the end of that decade the benefits of the railway technology began to feed through to the wider economy, several other trends including sovereign debt moved in a positive direction at the same time, and the result was startling economic growth. The post-war experience is the second example: the US emerged from the Second World War with debt above 100% of GDP and then delivered thirty-five years of deleveraging alongside relatively strong growth.

Figure 6: US debt-to-GDP and GDP growth rate



Source: Fineon, Bloomberg Finance L.P., Haver Analytics, Factset, Deutsche Bank Research

The lesson from both episodes is that high debt can be overcome when the other megatrends pull in the same direction. After WWII, globalisation expanded quickly, domestic politics were comparatively stable, social discontent was low, and energy technology was a net positive. The debt burden was carried by other things going well.

Today, things are different. Technology is positive force on the medium-term outlook for the global economy and energy modestly so. However, the remaining four megatrends — sovereign debt, demography, domestic politics and geopolitics — are having a negative effect. Technology, therefore, has a lot of work to do.

What this means for portfolios

We would draw four conclusions for medium-term positioning.

1. Investors spent Friday trading the Fed but the more durable question is what compensation is required to fund a debt stock heading towards 120% of GDP, especially if the average person continues to see equities as more attractive than bonds, even in bad times.
2. Bonds and (safe) currencies can be risk assets at different times: We have flagged before that haven assets could become risk assets, and that treasuries and the dollar in particular will become more volatile if the sovereign deficits megatrend deteriorates further. The haven correlation data since 2020 shows that hedging becomes materially harder when the hedge is itself exposed to the trend being hedged.
3. Sovereigns are increasingly being judged on demographics and fiscal credibility: Yields of countries with ageing populations and no credible fiscal plan for the resulting dependency ratios will that come under increased volatility.

4. If AI delivers, the sovereign debt problem becomes manageable and the equity market carries the benefit. If it does not, there is no obvious hedge. That argues for genuine exposure to the productivity outcome, but also for scepticism about business models that require capital intensity to keep rising indefinitely. Warsh's capital-light question cuts both ways.

What we are watching

Three key sovereign deficit variables we are tracking (among others):

1. The contribution of government spending to GDP. A potential bright spot: this indicator has recently risen sharply. If that continues alongside greater AI adoption, there is a path to sovereign deficits becoming less of a drag.

2. The US 10-year yield versus the G7 average. Treasury yields have been higher than the G7 average for most of this century, but the gap has begun to shrink. Continued convergence would signal that global demand for treasuries remains strong, which in turn supports the resilience of the US twin deficits.

3. The fiscal policy uncertainty index. Turned deeply negative in 2024 and 2025, below the Covid lows. Subjective, but an important lead indicator for short-term market views — particularly during an exogenous shock, where it is the transmission channel into fear-based sell-offs.

Appendix 1

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China - Modernizing the monetary policy framework

August 26, 2026

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Executive Summary

The **path to modernizing China's monetary policy framework has reached an important point.** The announcement by PBoC Governor Pan Gongsheng in June, introducing overnight reverse repo operations and stating the intent to narrow the interest-rate corridor, is a key step in this journey. **What is the ultimate destination, and what else does the PBoC need to do to get there?**

China currently operates a **hybrid monetary-policy framework** in which both policy rates and monetary/credit aggregates play important roles. The objective is ultimately to move away from a framework where monetary/credit aggregates serve as intermediate policy targets toward one where policy rates increasingly serve as the primary policy signal, with liquidity-management tools continuing to support policy implementation and transmission.

Key takeaways:

- Over the past decade, China's monetary policy framework has evolved toward greater use of price-based tools, although quantitative measures continue to play an important role.
- The objective is for policy rates to play a larger role in signaling the monetary stance, while liquidity-management and structural tools help with policy implementation.
- Key reform priorities include strengthening policy expectations, reinforcing the interest-rate corridor, deepening bond-market liquidity, improving credit transmission and increasing exchange rate flexibility.
- Improving the monetary policy framework alone may not be enough. Achieving more effective economic management will also require supportive fiscal policies, better coordination across policymakers and timely measures to stabilize growth.

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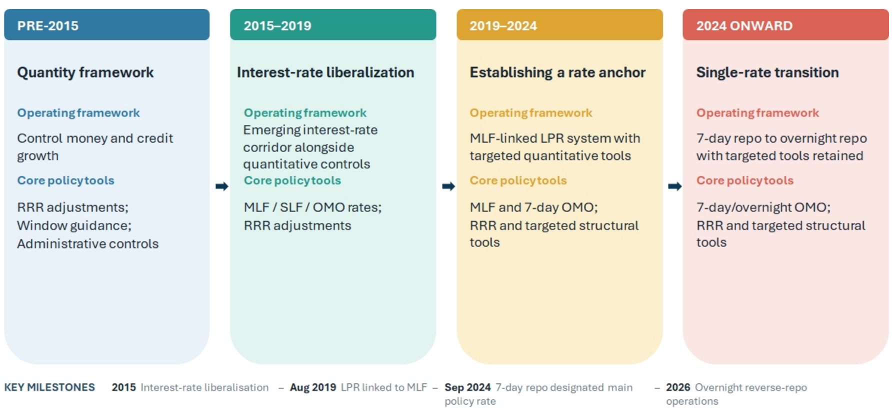
A decade of monetary policy modernization

Over the past decade, China has undertaken one of the most important monetary policy framework reforms among major economies. As interest-rate liberalization progressed, financial markets deepened and the economy became more complex, a framework centered mainly on quantitative controls became less effective. Policymakers therefore began shifting toward a more market-based system in which prices, rather than administrative controls, play a larger role in transmitting monetary policy.

Figure 1: China has gradually strengthened its price-based monetary policy framework over the past decade

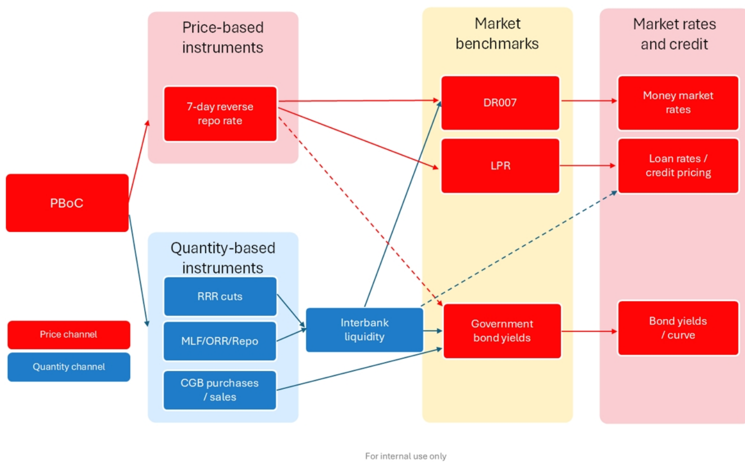


Source :Deutsche Bank Research

China's framework today looks much more interest-rate-based than it did a decade ago. Benchmark lending and deposit rates have largely been liberalized, the Loan Prime Rate has become a more important reference for bank lending rates, and the 7-day reverse repo rate now serves as the primary policy rate. At the same time, **this remains a hybrid system**. Today, the PBoC's policy toolkit can be broadly divided into four categories:

- **Policy rate**: the 7-day reverse repo rate serves as the primary policy rate and anchors short-term money-market rates.
- **Short- to medium-term liquidity tools**: Reverse repos, outright reverse repos (ORR), and the Medium-Term Lending Facility (MLF), which influence interbank liquidity conditions and bank funding costs.
- **Long-term liquidity tools**: Reserve requirement ratio (RRR) adjustments and outright purchases or sales of government bonds, which affect the structural liquidity position of the banking system.
- **Structural monetary policy tools**: Relending, rediscounting and other targeted facilities designed to channel credit towards strategic sectors and policy priorities (see Appendix B for the full list of liquidity tools).

Figure 2: China's monetary framework remains a hybrid system in transition



Source : Deutsche Bank Research

Some of the key building blocks of a modern interest-rate framework are now in place, with the 7-day reverse repo rate serving as the primary policy anchor and market-based interest rates playing a larger role in policy implementation than in the past. **The objective of China's monetary policy modernization is not the elimination of quantitative or structural tools, but the establishment of interest rates as the primary mechanism through which monetary policy influences financial conditions and economic activity.**

Figure 3: China's monetary policy differs from that of DMs more in objectives and implementation than tools

Dimension	PBoC	Fed	ECB	BoE	BoK
Primary objective	Price stability, growth support, credit conditions and financial stability all play important roles.	Inflation and employment	Price stability	Price stability with support for growth/employment	Price stability
Policy toolkit	Policy rate and liquidity operations	Policy rate and balance sheet	Policy rate and balance sheet	Policy rate and balance sheet	Policy rate and liquidity operations
No. of annual public communication	Quarterly monetary policy reports and occasional official speeches	Minutes and votes published, over 400 official speeches per year	Meeting accounts published, over 400 official speeches per year	Minutes and vote splits published, over 100 official speeches per year	Minutes and voting records published, ~10 official speeches per year
Communication style	Flexibility-focused. Quarterly reports and qualitative guidance	Highly transparent and forecast-driven	Rules and forecast-based	Forecast and minutes-driven	Relatively transparent by Asian standards
Forecast publication	Limited forecast disclosure	SEP projections	Staff forecasts	Detailed MPR forecasts	Growth and inflation forecasts
FX consideration	Exchange-rate stability remains an important consideration	Little direct role	Limited	Limited	Important but secondary

Source :Deutsche Bank Research; Note: Though financial stability is an important consideration for all central banks, it is not treated as a primary objective here due to the lack of an explicit and measurable target

Against this backdrop, an important question is whether the modernization of the monetary-policy framework has translated into stronger policy transmission. While there are signs of progress, the evidence remains mixed. Since January 2022, three of the seven policy rate cuts have been followed by a meaningful decline in lending rates within the subsequent month.

As a result, the next phase of monetary policy modernization is likely to focus on strengthening the channels through which policy rates influence financial markets and economic activity. This suggests that future reforms are likely to center on areas such as expectations formation, financial-market depth, interest-rate pass-through and the broader role of market signals within the monetary policy framework.

The next chapter of monetary policy modernization

What is China's likely destination?

The evidence above suggests that China's monetary-policy framework has entered a new phase of development. The key question for the next stage we believe is therefore not whether China will reduce its reliance on liquidity-management tools, but whether policy rates will increasingly displace monetary and credit aggregates as the primary guide for monetary-policy implementation and transmission. We believe the answer is yes. Over the past decade, policymakers have established many of the key building blocks of a more market-based framework, including a clearer policy-rate structure, greater reliance on market pricing and a reduced emphasis on monetary aggregates. As a result, the next phase of reform is likely to focus less on creating new instruments and more on strengthening the channels through which policy signals affect financial markets and economic activity.

Importantly, this does not imply convergence towards a conventional developed-market model. Structural and quantitative tools are likely to remain part of the framework, while financial stability, exchange-rate stability and development objectives will continue to influence policy decisions. Rather, modernization is likely to involve gradually increasing the role of interest rates as the primary mechanism through which monetary policy influences financial conditions, while improving the effectiveness of policy transmission within China's broader policy framework.

Moreover, China's evolving macroeconomic environment may reinforce the need for quantity-based and structural policy tools. As policy rates remain structurally lower amid accommodative policy stances, China may increasingly face constraints similar to those experienced by other major economies near the effective lower bound. In such an environment, conventional interest-rate adjustments may become less powerful, increasing the importance of balance-sheet policies, liquidity facilities and targeted credit-support measures. Rather than disappearing, quantity-based tools may therefore become a permanent complement to interest-rate policy, much as they have in many developed economies since the Global Financial Crisis.

Priorities for the next phase of monetary policy modernization

Simplifying policy objectives to strengthen market expectations. As China's monetary policy framework continues to evolve, policy communication and market expectations are likely to play an increasingly important role in shaping financial conditions. Modern monetary policy increasingly operates through expectations of future policy settings rather than current policy rates alone and the central of that is a clear set of policy objectives by the central bank. In China, the PBoC operates under a broad set of objectives, including growth, price stability, financial stability, exchange-rate stability and broader development goals, providing policymakers with flexibility. Continued development of the framework and clear communication of policy intentions could further support the role of expectations in financial markets.

Greater clarity around policy objectives would strengthen the expectations channel. This would not require adopting a formal inflation-targeting regime or a Fed-style dual mandate. A more pragmatic approach could be to streamline the PBoC’s mandate around three primary objectives. While the relative emphasis differs across jurisdictions, most major central banks operate with a similar set of objectives including inflation, growth or employment as well as financial stability.

Aligning the PBoC's mandate more explicitly around these objectives could improve policy transparency without requiring a wholesale adoption of a Western monetary-policy framework. A more focused mandate, supported by clear and consistent communication, would make the PBoC's reaction function easier for markets to interpret and enhance the signaling power of policy-rate changes.

Strengthening the corridor to improve short-term transmission. A second priority is improving the operating framework itself. China has made considerable progress in establishing the 7-day reverse repo rate as the primary policy rate, while Governor Pan has also signaled a possible future transition towards a more overnight-rate-based framework. At the same time, the interest-rate corridor continues to evolve as policymakers refine the monetary-policy operating framework. Although temporary repo and reverse repo operations were introduced in 2024 and reformed in 2026, they have not yet become a significant part of day-to-day operations. As a result, their role in guiding money-market conditions and supporting policy implementation will remain an area of interest for market participants.

In the United States, market rates such as SOFR generally trade close to policy rates within a well-established floor system. By contrast, China's corridor remains more of an evolving guide than a tightly binding operating framework. Future reforms are therefore likely to focus on narrowing the corridor, strengthening overnight liquidity operations and improving the predictability of short-term liquidity management.

A further area of development is the broader financial-stability architecture underpinning monetary operations. Recent discussions around emergency liquidity facilities for non-bank financial institutions suggest policymakers increasingly recognize the need for stronger liquidity backstops as financial markets become more market-based. A more comprehensive liquidity-safety framework could improve market resilience and support the implementation of monetary policy during periods of stress.

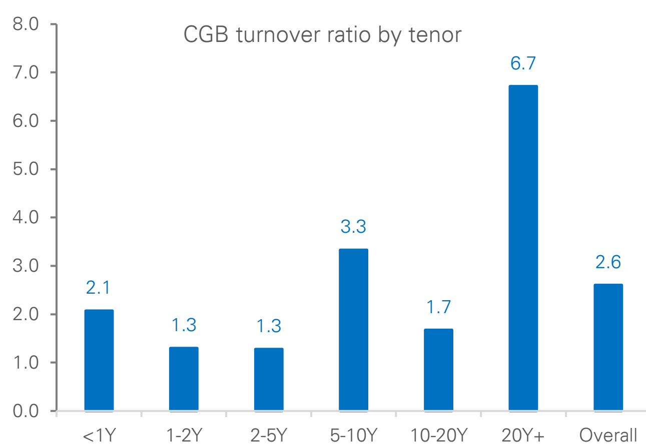
Deepening secondary bond markets to improve yield-curve transmission.China has built one of the world's largest government bond markets over the past two decades. While commercial banks continue to hold a significant share of outstanding government bonds, policymakers have also encouraged broader participation by institutional and foreign investors. As secondary-market activity continues to evolve, deeper liquidity and more active trading could further enhance the informational content of market interest rates and the role of the yield curve within the broader monetary-policy framework.

A deeper and more active government bond market would strengthen the transmission of policy expectations along the yield curve. Potential reforms include broadening participation beyond commercial banks, developing government-bond derivatives and hedging markets, improving repo and securities-lending infrastructure, strengthening market-making arrangements and enhancing transparency around issuance and liquidity operations.

These reforms would improve monetary policy transmission while strengthening the role of government bond yields as benchmarks for broader financial-market pricing.

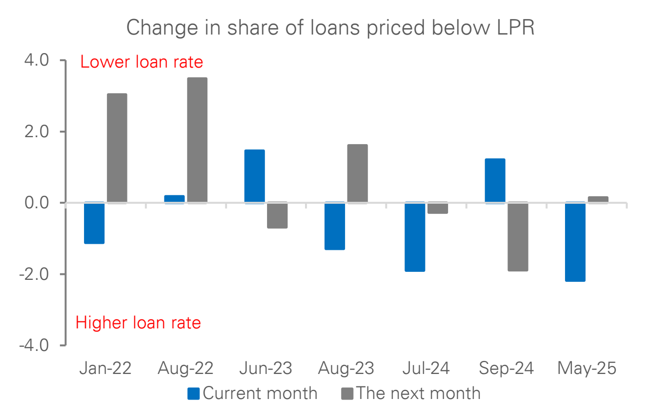
Improving liquidity circulation and credit transmission. Another area of development relates to the credit channel and the interaction between financial conditions and economic activity. While policy easing has supported ample liquidity conditions in recent years, the impact on borrowing, investment and growth can vary depending on broader economic and business conditions. China's financing system remains predominantly bank-based, with around 80% of corporate financing coming from loans, a higher share than in many major economies. Policymakers have therefore placed increasing emphasis on broadening financing channels and supporting the development of debt and equity markets alongside traditional bank lending. Consistent with this objective, PBoC Governor Pan Gongsheng in 2023 highlighted efforts to expand financing channels for private enterprises across debt, equity and bank lending. Going forward, the evolution of credit transmission is likely to reflect not only monetary-policy developments, but also broader economic conditions and structural reforms.

Figure 4: A more active bond market could support price discovery and transmission



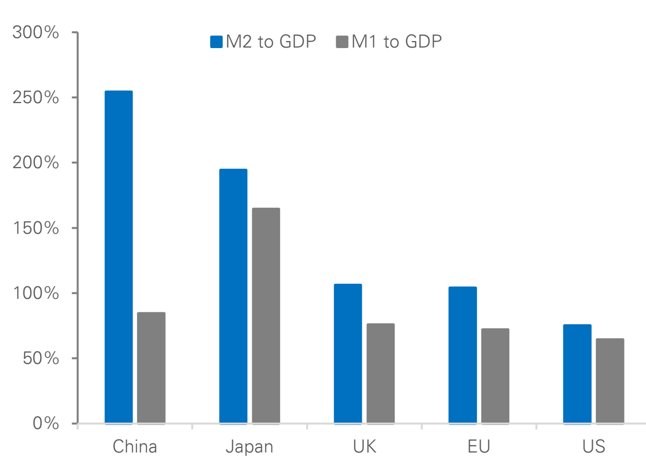
Source : Deutsche Bank Research, Wind

Figure 5: Policy-rate cuts have not always been followed by immediate declines in loan rates



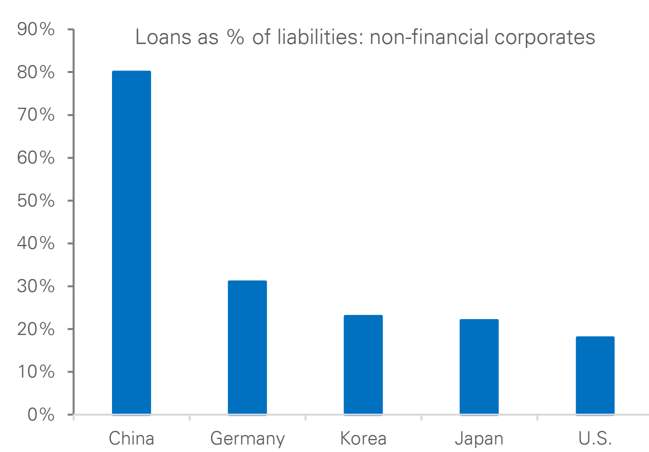
Source : Deutsche Bank Research, Wind

Figure 6: Liquidity and economic activity can diverge over time



Source : Deutsche Bank Research, Wind

Figure 7: Chinese corporates remain heavily reliant on bank-loan financing



Source : Deutsche Bank Research

Exchange-rate developments and RMB internationalization. Another area of development concerns the interaction between monetary policy, exchange rates and international financial flows. China's monetary-policy framework operates within a broader set of policy objectives, resulting in transmission dynamics that differ from those in many developed-market economies. As RMB internationalization advances and financial-market development continues, the relationship between monetary policy, capital flows and exchange-rate developments is likely to evolve further. This process will remain an important area to watch as China's financial system becomes increasingly integrated with global markets.

A modern framework, not a Western one. Ultimately, in our view, monetary policy modernization in China should not be viewed as a process of convergence towards a developed-market model. Rather, it is a process of adapting market-based monetary policy tools to China's own institutional environment. The likely end state, we believe, is a framework with a clear policy rate, deeper and more responsive financial markets, stronger policy transmission and greater transparency, while retaining broader policy objectives and strategic credit-support mechanisms. In other words, China is unlikely to become a Chinese version of the Federal Reserve. Instead, we think it is more likely to develop a modern interest-rate framework with distinctly Chinese characteristics.

Beyond monetary policy reform: A broader policy agenda

The modernization of China's monetary policy framework has brought it closer to the operational practices of many developed-market central banks. The PBoC has increasingly emphasized price-based policy instruments, strengthened its policy-rate framework, and improved the signaling and implementation of monetary policy. These developments represent an important step in the evolution of China's monetary-policy framework and should help further strengthen the role of market-based interest rates over time.

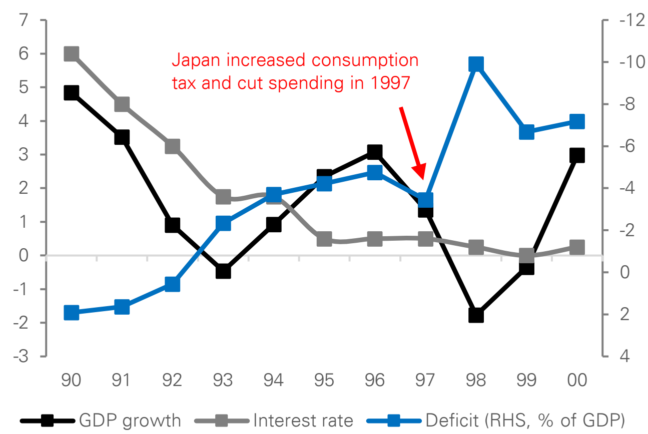
At the same time, monetary policy operates within a broader economic and financial environment. Factors such as credit demand, developments across different sectors of the economy, and ongoing adjustments in the property market can all influence how monetary-policy actions are transmitted through the economy. As a result, the evolution of monetary-policy tools is only one part of the broader policy landscape. Experience from developed economies similarly suggests that monetary policy is often most effective when supported by favorable macroeconomic and financial conditions.

More broadly, China's economic objectives are pursued through a combination of monetary, fiscal, regulatory and structural policies. We therefore view monetary-policy modernization as one component of a broader policy agenda rather than a standalone initiative. Going forward, the interaction between monetary policy and other policy measures will remain an important factor shaping economic and financial outcomes.

Monetary policy and fiscal policy interaction. The effectiveness of monetary policy is often influenced by the broader macroeconomic policy environment, including fiscal policy. Japan's experience in the late 1990s illustrates the importance of policy interaction. Despite very low interest rates and repeated monetary easing, fiscal consolidation in 1997 coincided with renewed economic weakness and a prolonged period of low inflation.

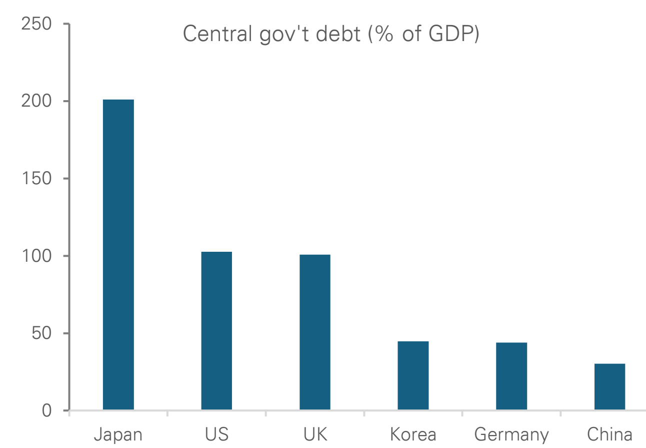
In China's case, sustaining monetary policy modernization, and potentially an easing cycle in the coming years, will need more forceful fiscal support. In recent years, policymakers have gradually expanded fiscal support while emphasizing the role of the central government in supporting economic stability. With a comparatively strong central-government balance sheet and a low-interest-rate environment (Figure 13), fiscal policy retains scope to play an important role in supporting broader macroeconomic objectives. More broadly, the interaction between monetary and fiscal policy is likely to remain an important feature of China's macroeconomic framework in the years ahead. More proactive fiscal support could help stabilize the property market, strengthen the social safety net, and support household consumption. Importantly, stronger fiscal support would complement monetary easing by generating demand for credit and investment opportunities, thereby improving the effectiveness of monetary policy itself.

Figure 8: Japan shows uncoordinated policies weaken transmission



Source : Deutsche Bank Research, Haver

Figure 9: China's central-government balance sheet remains among the strongest in the world



Source : Deutsche Bank Research, Haver, Wind

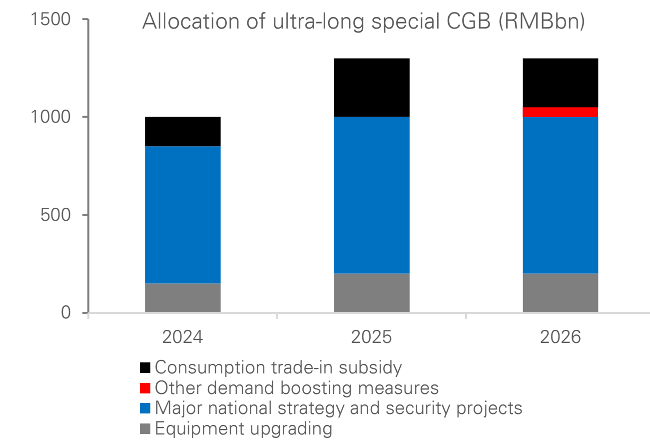
Better balancing long-term strategic priorities with near-term economic needs. China's policy agenda has increasingly emphasized long-term priorities such as technological innovation, advanced manufacturing and the development of future industries. Between 2023 and 2026, a significant share of funding raised through ultra-long-term special government bonds was directed toward these strategic objectives, highlighting policymakers' focus on enhancing long-term productivity growth and competitiveness.

At the same time, long-term development and cyclical stabilization can be mutually reinforcing. A stable macroeconomic environment can support employment, household income, business activity and investment, helping to create favorable conditions for longer-term economic upgrading. In this sense, cyclical conditions and structural transformation need not be viewed as competing objectives.

International experience suggests that strategic development policies and macroeconomic stabilization measures can coexist. South Korea, for example, has combined support for priority sectors such as semiconductors with broader macroeconomic measures aimed at maintaining economic stability during periods of cyclical weakness. More generally, a stable economic environment can provide an important foundation for investment, innovation and industrial upgrading over time.

Looking ahead, the interaction between China's long-term development objectives and broader economic conditions will remain an important theme for policymakers and markets. The pace of structural transformation, developments across key sectors of the economy, and broader macroeconomic conditions will all play a role in shaping the environment in which longer-term strategic goals are pursued.

Figure 10: Most incremental fiscal resources were allocated to strategic sectors



Source :Deutsche Bank Research, Government Work Report

Figure 11: Korea's experience shows that strategic upgrading and cyclical growth can go hand in hand

	China		Korea	
	Name	Amount	Name	Amount
Strategic and industrial policies	Major national strategy and security projects (2024–2026)	RMB2.3tn (1.6% of annual GDP)	K-Chips Act (2023)	Tax incentive (not direct spending)
	Policy financing tool to support specific strategic sectors	RMB1.3tn (0.9% of annual GDP)	Semiconductor Ecosystem Support Package (2024)	KRW26tn (1% of annual GDP)
			Chip fund (2026)	KRW5tn (0.2% of annual GDP)
Cyclical growth policies	Consumption trade-in subsidies (2024–2026)	RMB700bn (0.5% of annual GDP)	Budget expansion to boost economic activity (2025)	KRW20.2tn (0.8% of annual GDP)
	Childcare subsidies (2025–2026)	RMB180bn (0.1% of annual GDP)	Supply-shock stimulus (2026)	KRW17.3tn (0.6% of annual GDP)

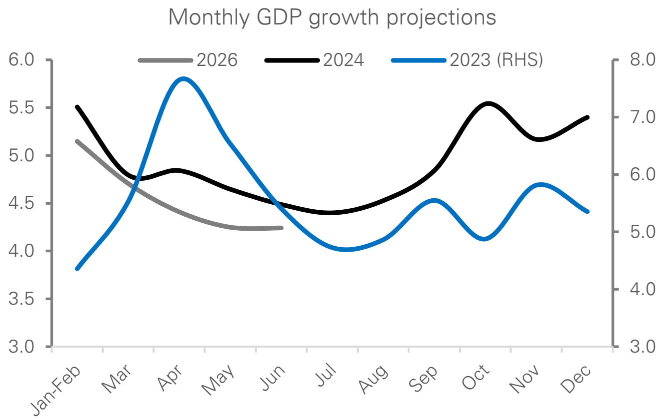
Source : Deutsche Bank Research
Note: China's strategic-sector allocation excludes investment funded by local government special bonds, which amount to around 3% of GDP annually

Proactive policy support and greater policy consistency. Expectations play an important role in shaping the effectiveness of macroeconomic policy. As such, the consistency and coordination of policy measures can influence household, business and market expectations, alongside broader economic outcomes. China's post-pandemic recovery has exhibited recurring fluctuations in growth momentum, with periods of policy support often coinciding with firmer economic activity. While policy measures have helped support activity, the consistency and persistence of policy support remain important considerations in anchoring expectations and strengthening confidence.

International experience also points to the importance of policy consistency. Japan's experience during the 1990s and 2000s demonstrated how expectations can be influenced by the interaction between monetary, fiscal and broader economic policies. Following 2013, a more coordinated policy framework was accompanied by improvements in growth, inflation and market sentiment, although external factors and global developments also played an important role.

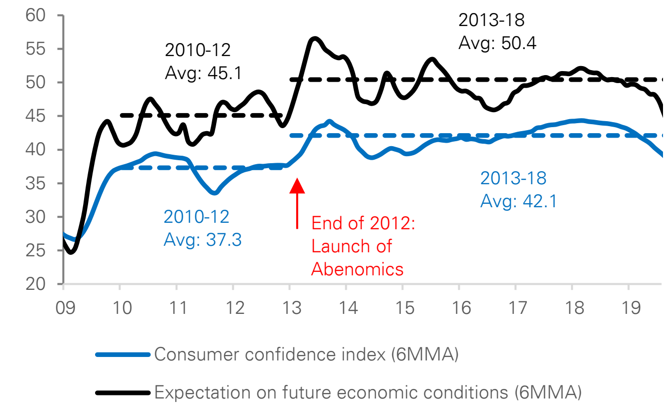
More broadly, we view the modernization of China's monetary-policy framework as an important enabling reform. A more market-based framework may further strengthen policy implementation and the role of interest rates within the financial system. At the same time, economic outcomes will continue to reflect a wide range of factors, including fiscal policy, structural developments and the broader macroeconomic environment. In this regard, the evolution of China's policy framework is likely to be shaped not only by monetary-policy reforms, but also by how different policy initiatives interact in support of longer-term economic objectives.

Figure 12: Growth cycles have highlighted the importance of policy consistency



Source : : Deutsche Bank Research, CEIC
Note: Monthly GDP growth is estimated by interpolating official GDP data based on our monthly nowcast.

Figure 13: Abenomics triggered an immediate and lasting recovery in consumer confidence



Source :Deutsche Bank Research, CEIC

Market Implications

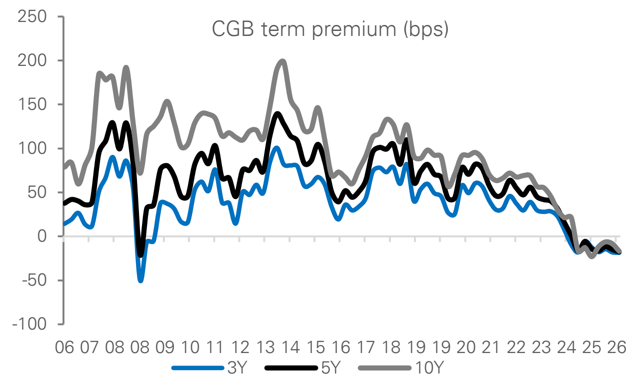
If China's monetary-policy reforms continue, they could gradually reshape the country's bond and currency markets. Over time, policy signals may become clearer, interest-rate changes may transmit more effectively through the financial system, and markets may become deeper and more responsive.

A steeper yield curve for China Government Bonds (CGBs). China's government-bond yield curve has become much flatter in recent years. This reflects both expectations that policy rates will remain low for some time and a decline in the extra compensation investors typically demand for holding longer-term bonds, known as the term premium. If monetary-policy modernization improves transmission and is supported by broader reforms, the yield curve could become steeper over time. Short-term yields could remain low if policy easing continues, while long-term yields could gradually rise as economic activity, confidence and inflation expectations strengthen.

Near-term easing, medium-term repricing higher for interest rate swaps (IRS). China's medium-term interest-rate expectations have moved lower in recent years as growth and inflation have moderated. Further policy easing could keep rates low in the near term. Over time, however, if reforms help support investment, lending and economic activity, expectations for future growth may improve. In that scenario, medium-term interest rates could gradually move higher and market pricing could become more balanced across different maturities.

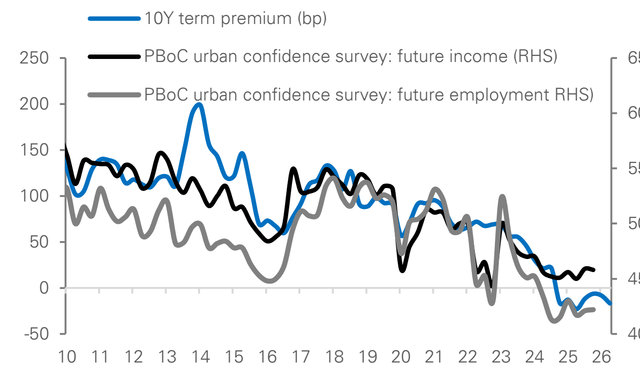
Greater volatility in RMB exchange rate. The RMB remains less volatile than many major currencies. If China continues modernizing its monetary policy framework and allows greater exchange-rate flexibility over time, the RMB could move more freely in response to economic conditions and market forces. This would likely result in greater two-way currency fluctuations while improving price discovery and supporting the development of deeper hedging and foreign-exchange markets. Given China's importance to the regional economy, greater RMB flexibility could also contribute to increased volatility in other Asian currencies that are closely linked to China through trade and financial flows.

Figure 14: CGB term premia have been declining across tenors



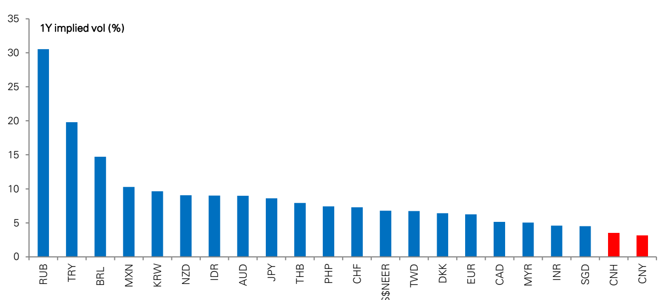
Source : Deutsche Bank Research, Bloomberg Finance LP

Figure 15: Stronger economic expectations could boost the term premium



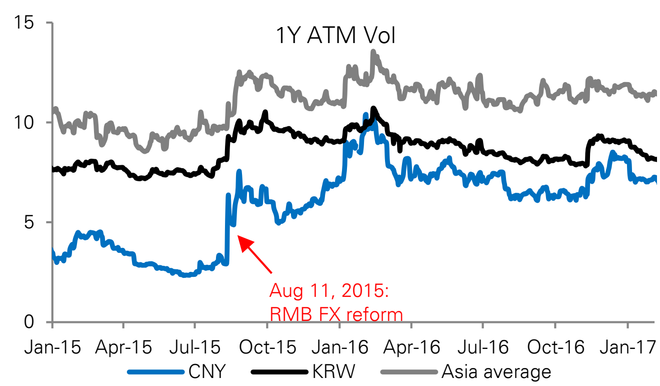
Source :Deutsche Bank Research, Bloomberg Finance LP, Wind

Figure 16: Rising RMB volatility could lift volatility across Asian FX markets



Source : Deutsche Bank Research, Bloomberg Finance LP

Figure 17: RMB FX regime change could have a significant impact on regional volatility



Source : Deutsche Bank Research, Bloomberg Finance LP

Appendix 1

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Deutsche Bank
Research

AI at 70: 14 lessons from a lifetime of boom and bust

August 2026

Adrian Cox

Deutsche Bank Research Institute

With deep dedication.

Deutsche Bank
Research Institute

IMPORTANT RESEARCH DISCLOSURES LOCATED IN APPENDIX 1.

Artificial intelligence marks 70 years since Dartmouth Summer Research Project

“Context is all you need”: clues from the past for what comes next

Seventy years ago today, the curtain came down on the summer workshop that gave birth to artificial intelligence. Dwight Eisenhower was in the White House, the Suez Crisis was gripping Britain and Elvis Presley topped the US charts with “Don’t Be Cruel” and “Hound Dog”.

The Dartmouth Summer Research project brought together a dozen or so academics convinced that every aspect of intelligence could in principle be described precisely enough for a machine to simulate it. Emboldened by the techno-optimism of the age, John McCarthy, who coined the name AI, wrote that “we think that a significant advance can be made… if a carefully selected group of scientists work on it together for a summer.”

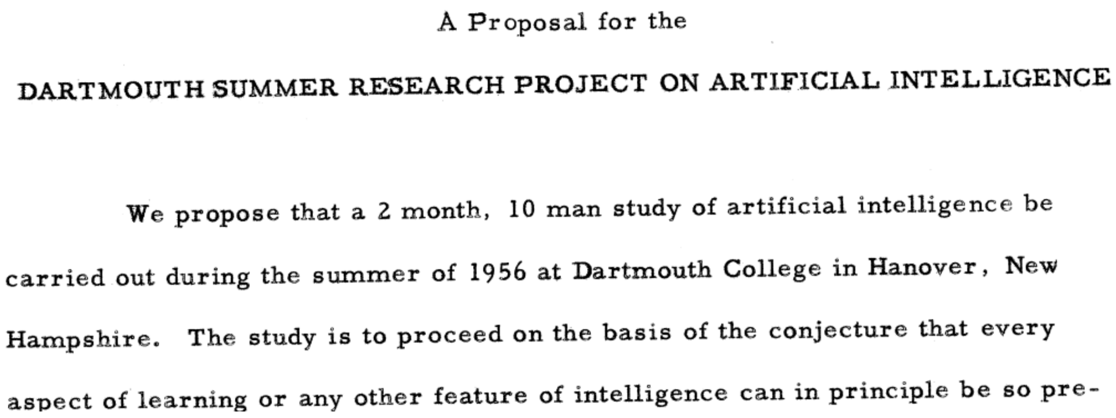
Since then, AI has lived a lifetime of summers and winters, punctuated by stunning breakthroughs and false dawns. As it is now the cornerstone of a new investment and valuation boom, the past 70 years are a treasure trove of clues for what to expect in the future.

To paraphrase the research paper that sparked the current boom in generative AI in 2017, “Attention is all you need”, you could say “context is all you need” to understand what is to come. The signals are there, from non-linear growth to infrastructure constraints and valuation booms and busts.

Some will say “this time is different”. Maybe. But the following charts give a flavour of what could happen if it isn’t. To quote the UK No. 1 single from 70 years ago, sung by Doris Day: “Whatever Will Be, Will Be (Que Sera, Sera)”.

14 lessons from a lifetime of boom and bust

1. AI progress is non-linear
2. Progress exceeds Moore's law
3. Today's leading tech may not be tomorrow's
4. The pipeline matters
5. Cheaper AI will likely boost spending
6. Software revolutions need hardware
7. Hardware is often a bottleneck
8. Globalisation has made tech cheaper, and more at risk
9. Tech revolutions take time to pay
10. Consumers have adopted AI faster than businesses
11. The biggest valuation booms are often tech booms
12. This time is different – and will need to be
13. Early leads do not guarantee long-term winners
14. Long-term trends tend to persist

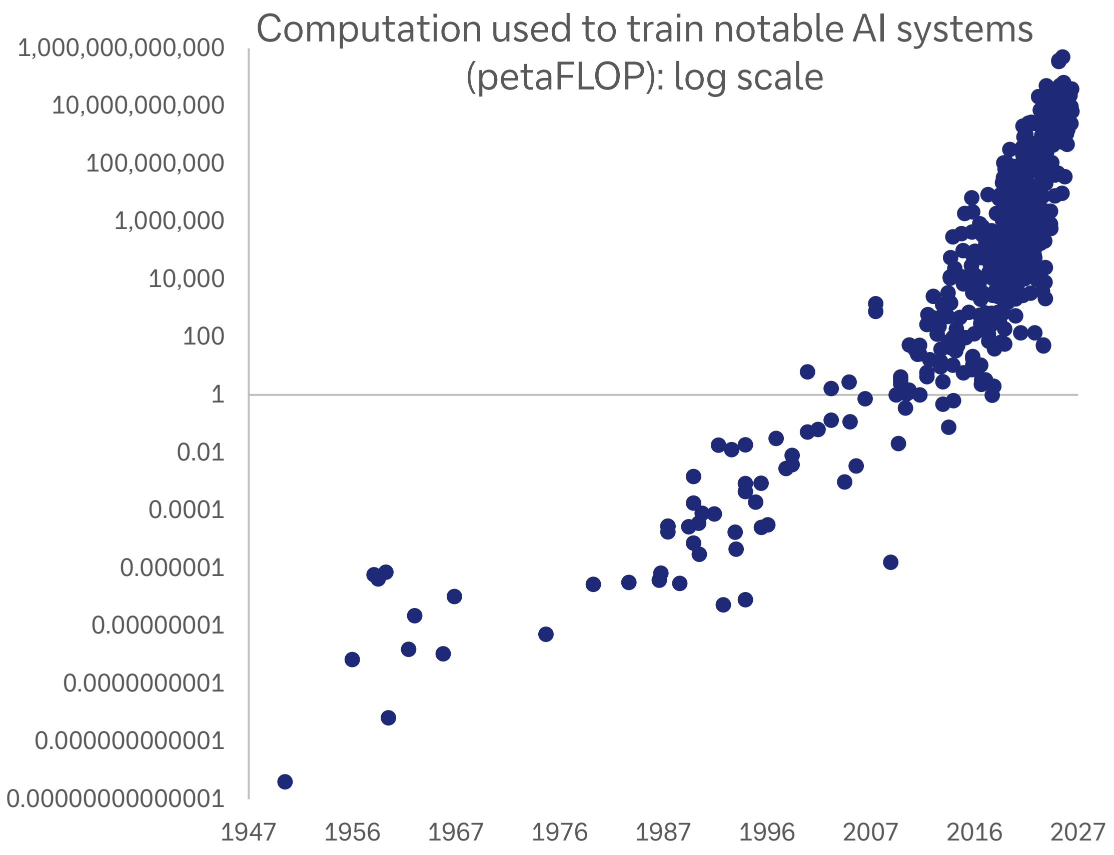

A Proposal for the

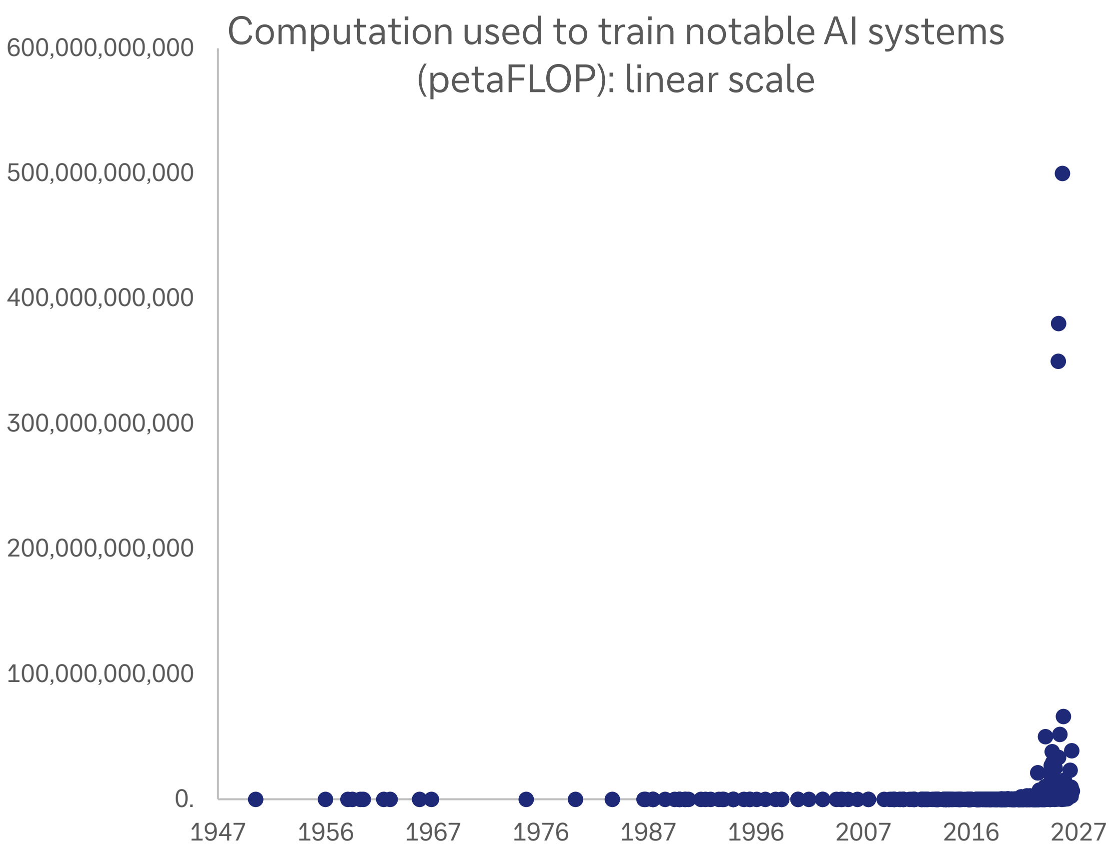
DARTMOUTH SUMMER RESEARCH PROJECT ON ARTIFICIAL INTELLIGENCE

We propose that a 2 month, 10 man study of artificial intelligence be carried out during the summer of 1956 at Dartmouth College in Hanover, New Hampshire. The study is to proceed on the basis of the conjecture that every aspect of learning or any other feature of intelligence can in principle be so pre-

1. AI progress is non-linear – and exponential growth is almost invariably underestimated

Compare what AI charts usually look like (on the left) vs what they actually represent (on the right)

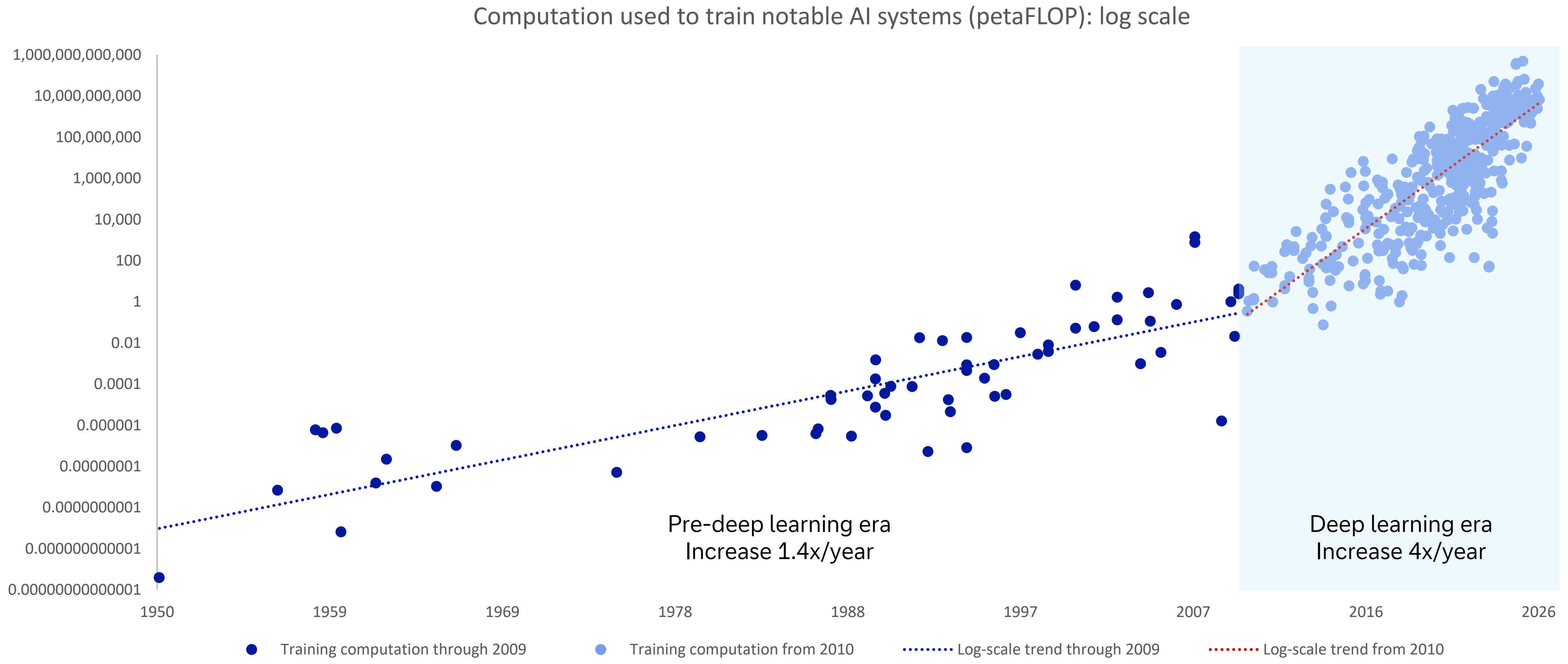




Source: Epoch AI (citing estimates from AI literature), Our World in Data, Deutsche Bank Research. Note: One FLOP (floating-point operation) represents a single arithmetic operation using floating point numbers; one petaFLOP is one quadrillion FLOPs

2. The pace of progress exceeds Moore's law when many factors are improving at once

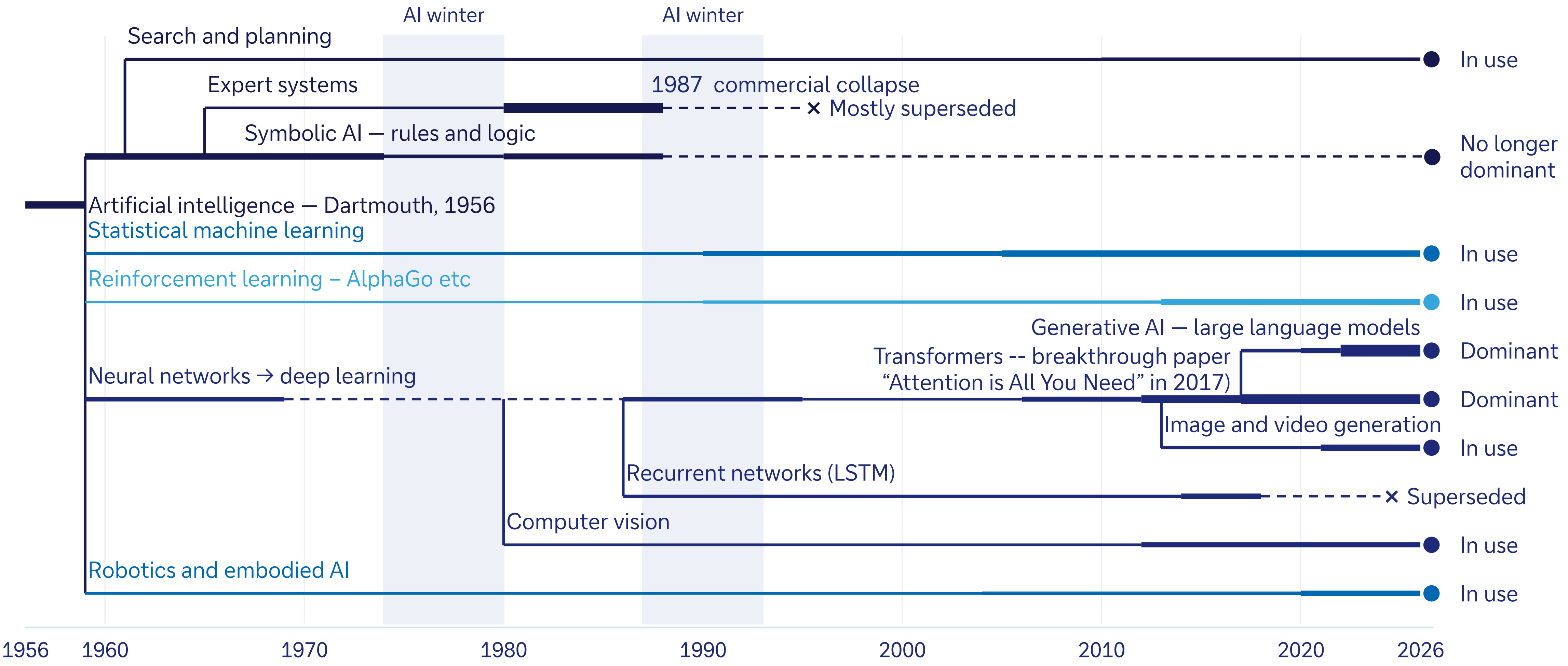
Conventional computing capability has doubled every 18-24 months but bigger and better systems, memory and algorithms have accelerated capabilities far beyond that



Source: Epoch AI (citing estimates from AI literature), Our World in Data, Deutsche Bank Research

3. Today's leading tech may not be tomorrow's

AI has an ever-evolving family tree; large language models may give way to, eg, world models



Colour = lineage · Thicker line = more activity · Shaded = AI winters · × = superseded · ● = still in use

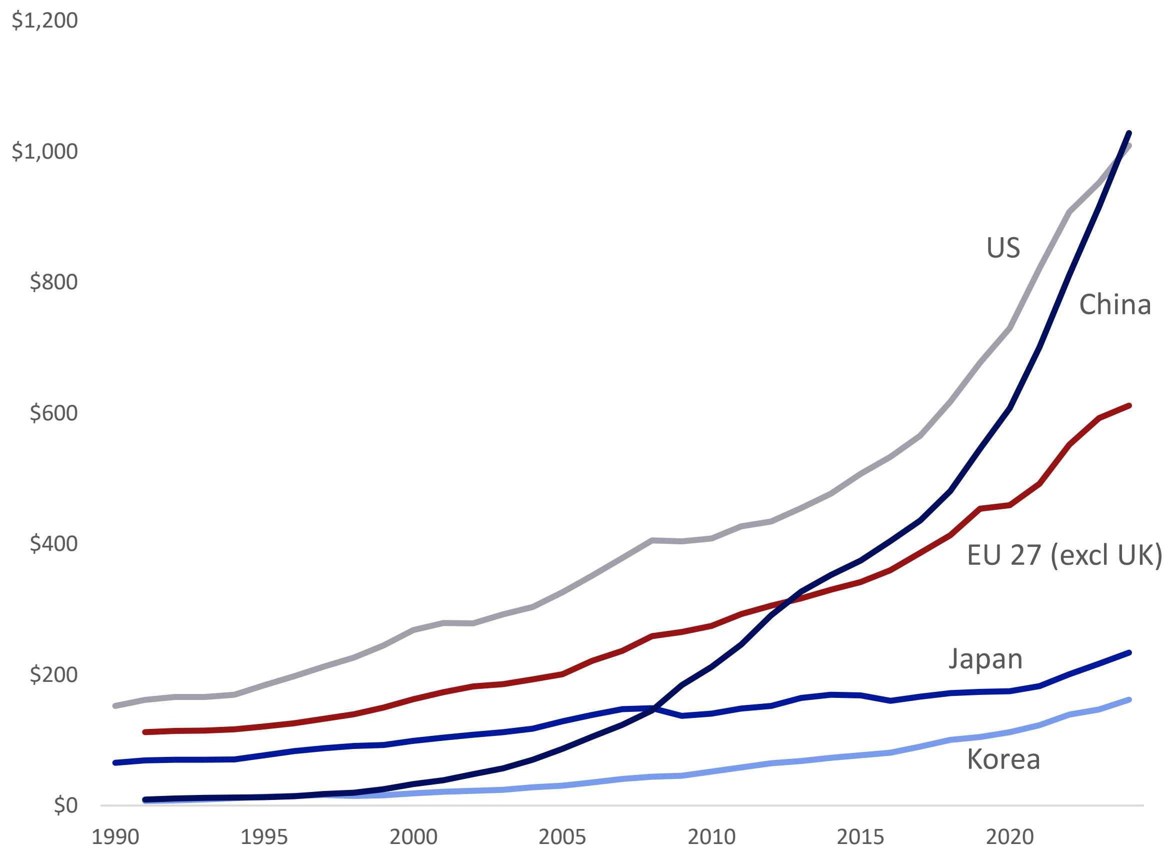
Dates mark impact, not first invention.

Source: Deutsche Bank Research

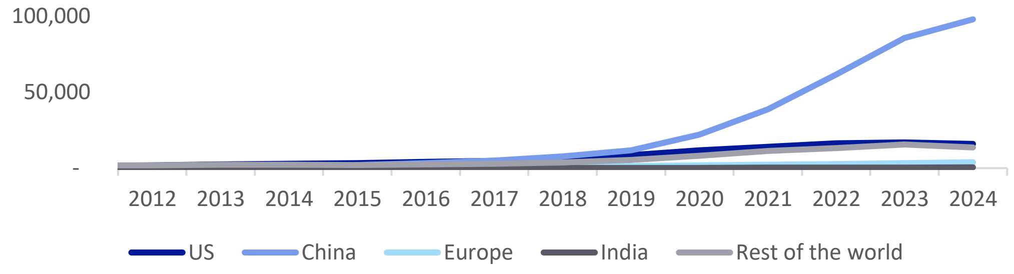
4. The pipeline matters: China's overnight success with DeepSeek was years in the making

Success of Chinese models reflects a multi-year push in R&D, overtaking the US in 2024

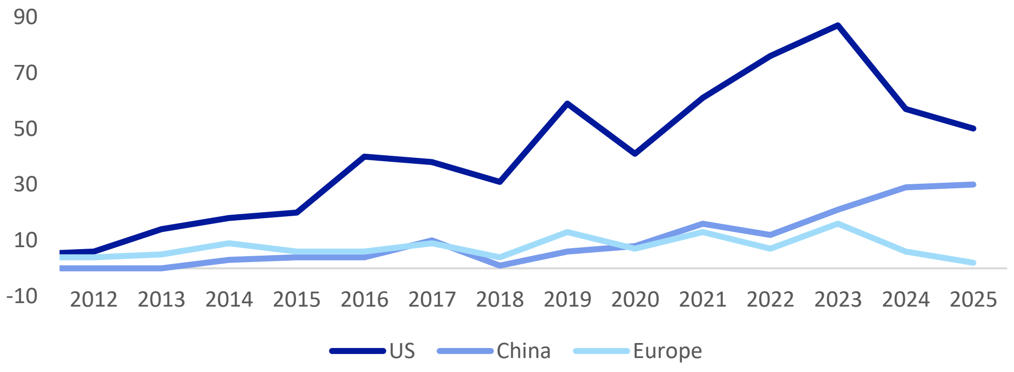
Gross domestic research and development spend by country, PPP converted, current prices (USDbn)



Number of AI patents granted by select geographic areas, 2012-2024



Number of notable AI models by geographic area (2012-2025)

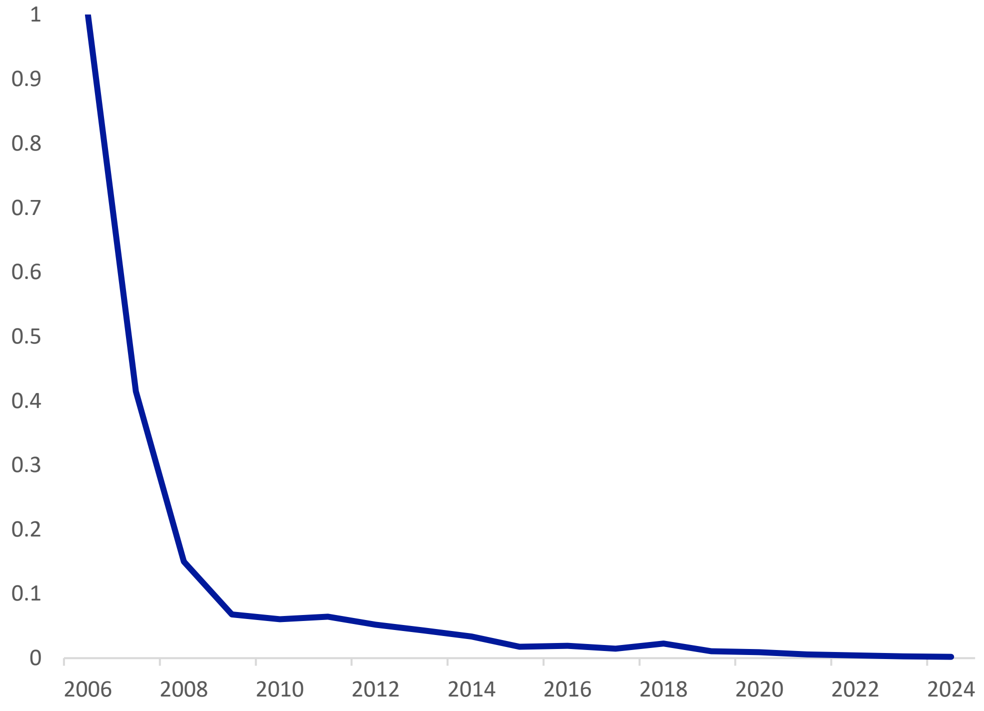


Source: OECD Data Explorer, Deutsche Bank Research

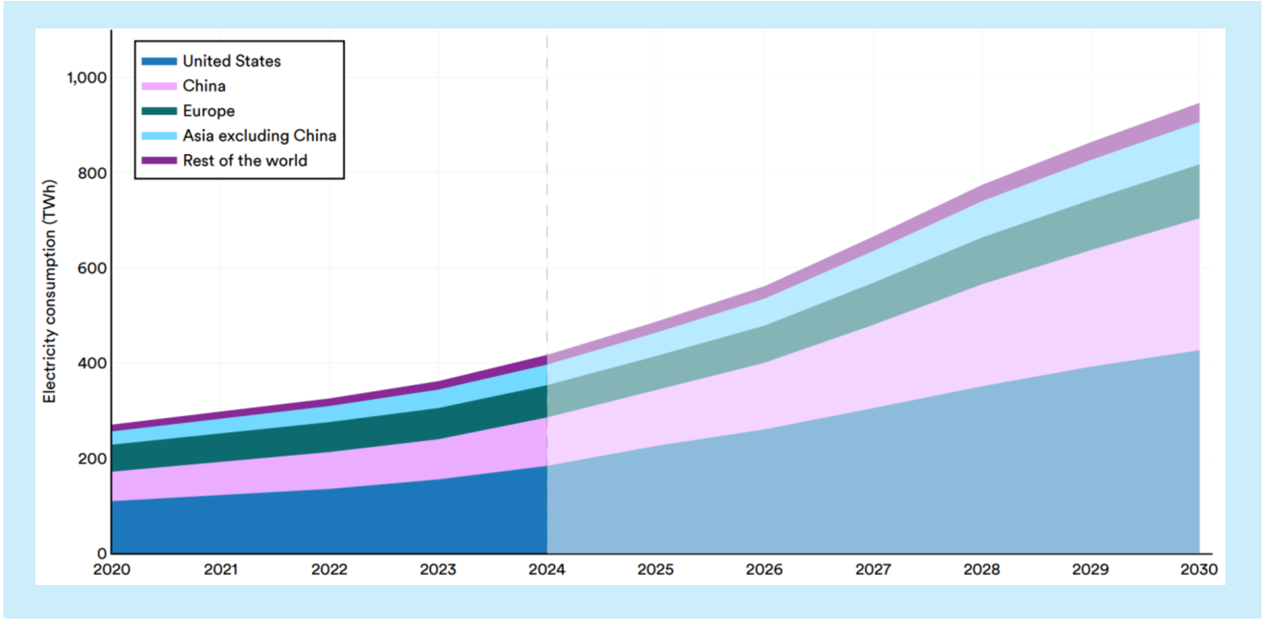
5. Making AI dramatically cheaper will likely increase, not decrease, spending

Jevons Paradox is alive: GPU computation cost has fallen by more than 99% since 2006, yet electricity consumption by data centres is set to double from 2024 to 2030 as demand surges

GPU chip computation cost index (2006=1)



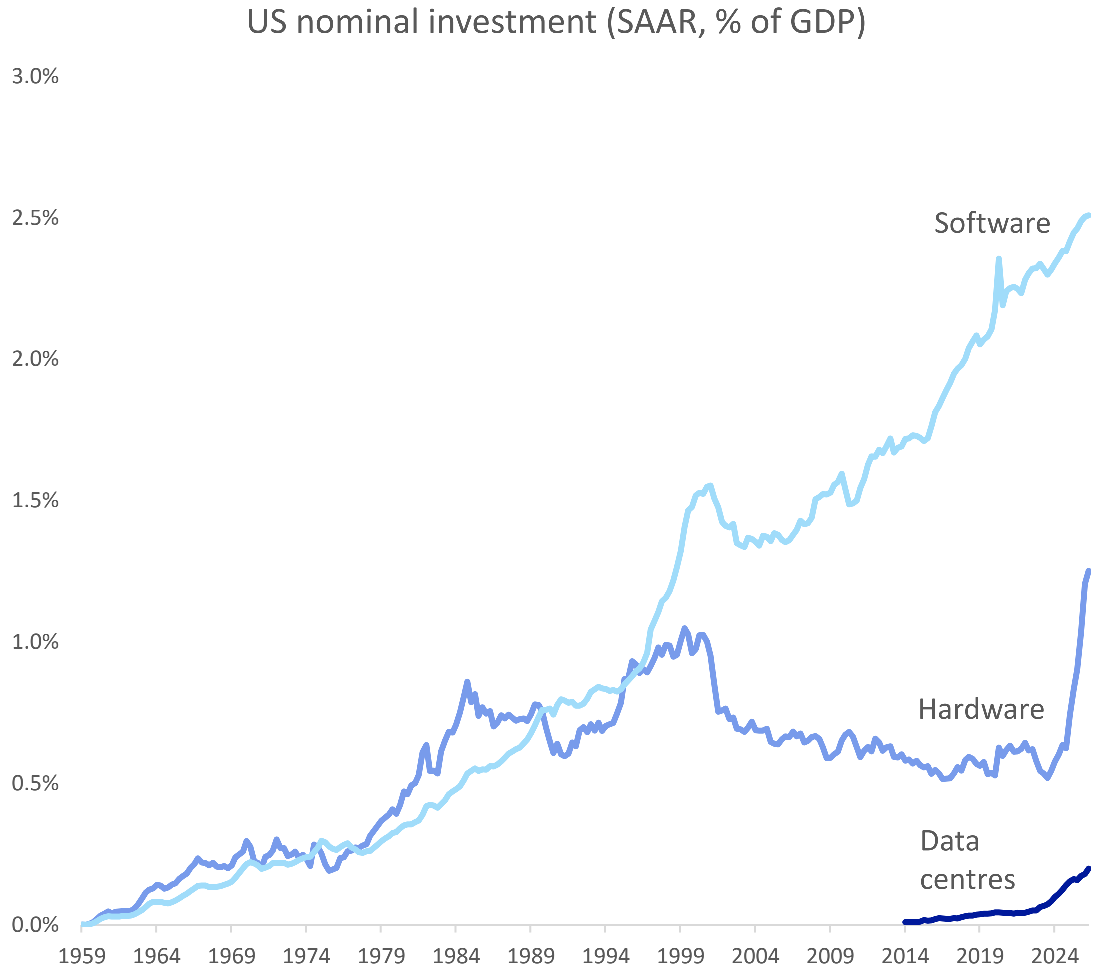
Data centre electricity consumption by region 2020-2030 (IEA projections)

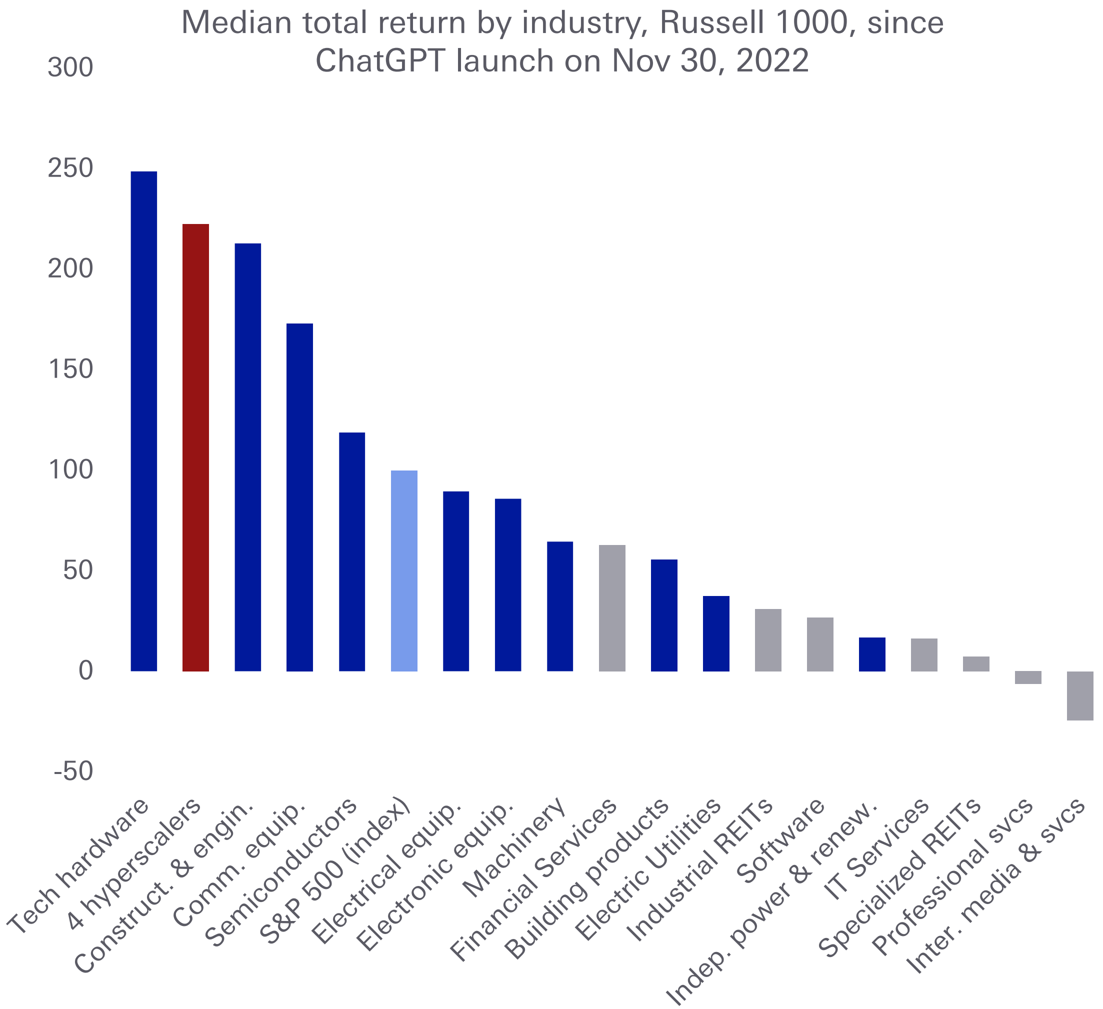


Source: International Energy Agency (2025), Stanford HAI AI Index 2026, Deutsche Bank Research

6. Software revolutions depend on heavy hardware investment

Hardware, construction, chip and equipment makers have led gains in the AI boom so far



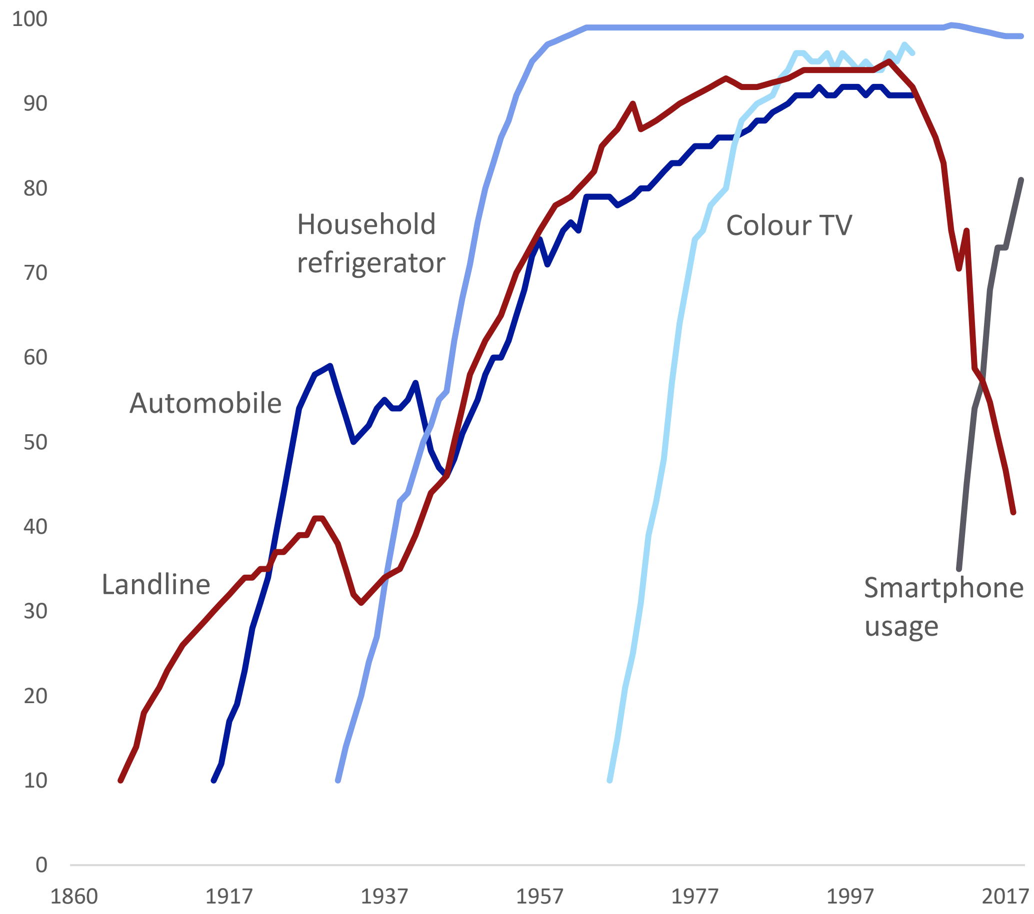


Source: Federal Reserve Bank of St Louis, Haver Analytics, Bloomberg Finance LP, Deutsche Bank Research

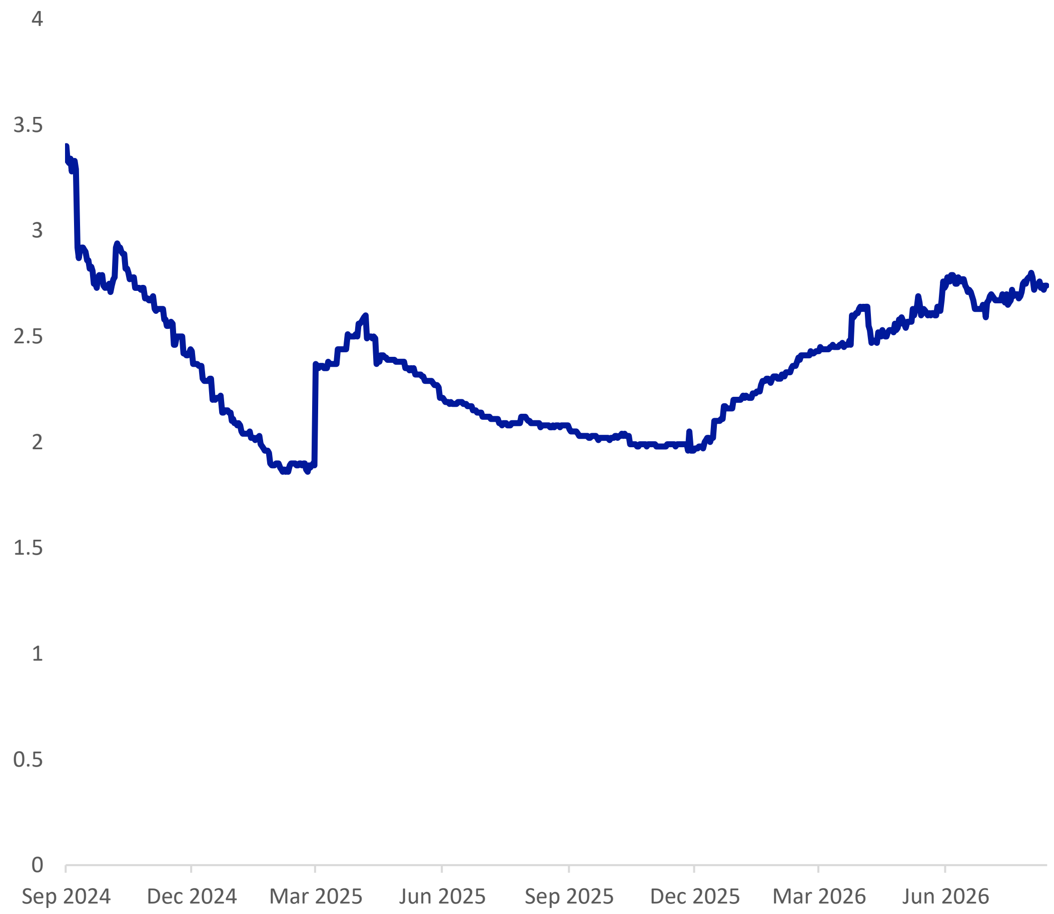
7. Hardware is often a bottleneck – but for once the delay is with producers not consumers

Past booms often relied on getting devices to consumers; now it's about getting scarce hardware online to serve them remotely

Share of US households using specific technologies (%)



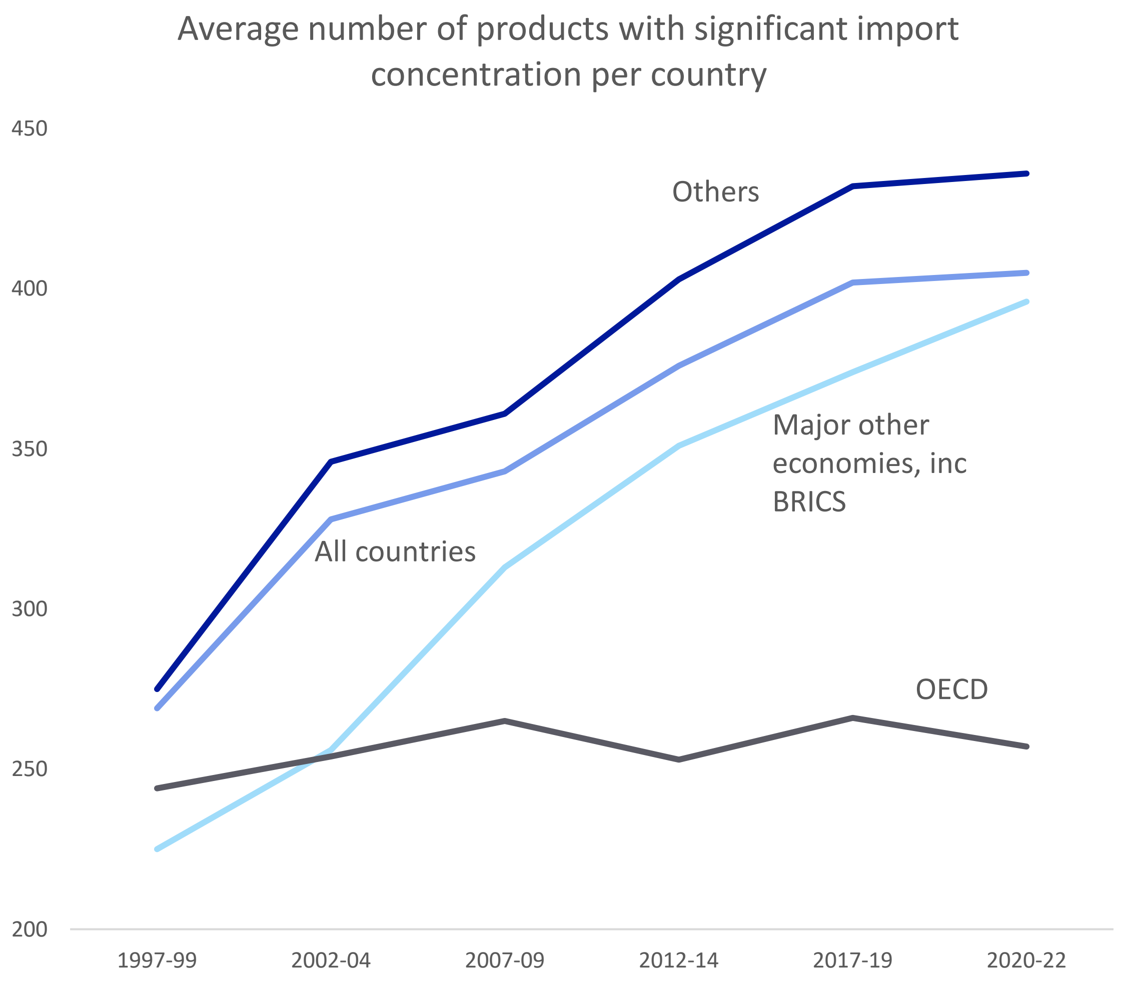
Hourly rental price of Nvidia H100 GPU chips (USD)

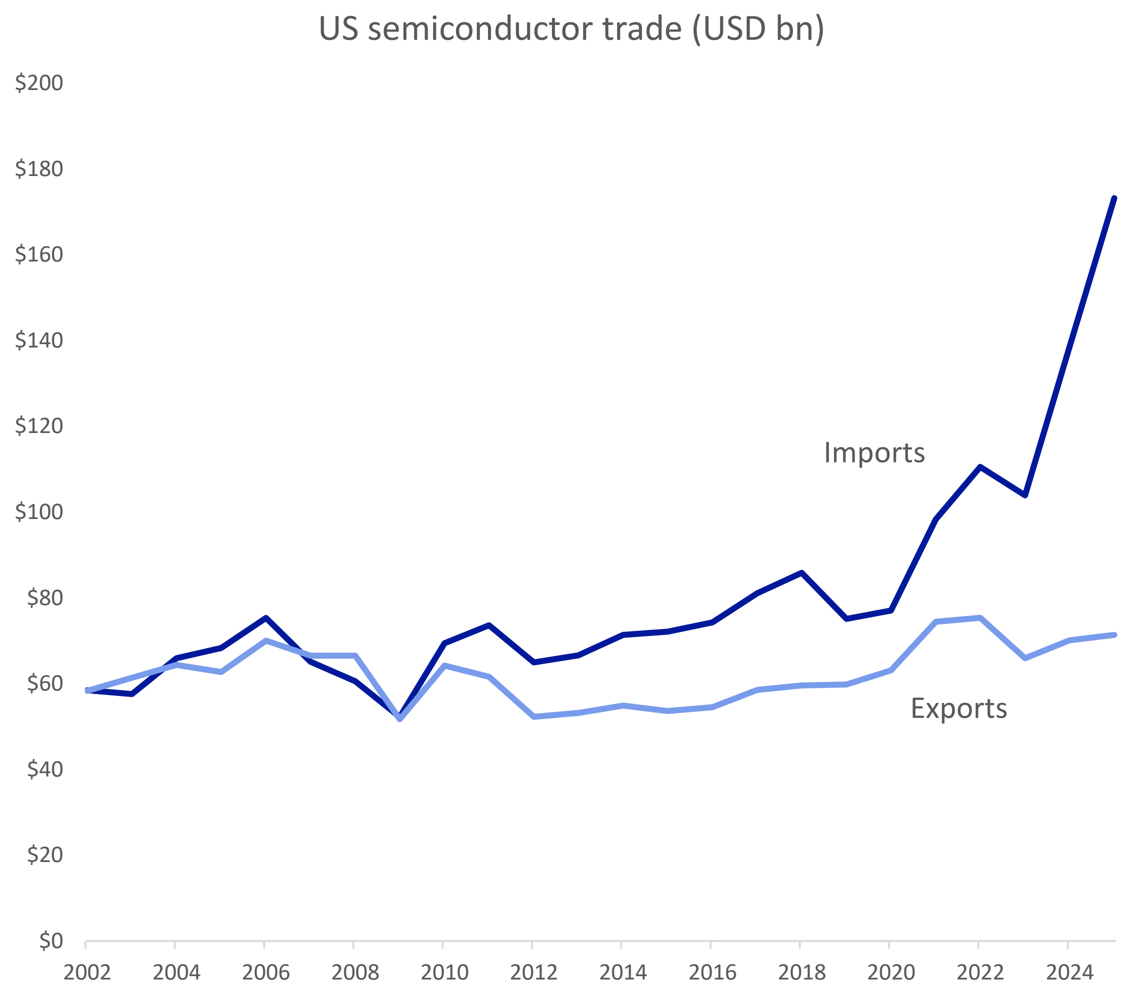


Source: Our World in Data, Federal Reserve Bank of St Louis, Haver Analytics, Bloomberg Finance LP, Deutsche Bank Research

8. Globalisation made tech cheaper – but also increased supply chain risks

There has been a surge in US imports of semiconductors as the AI boom has taken off

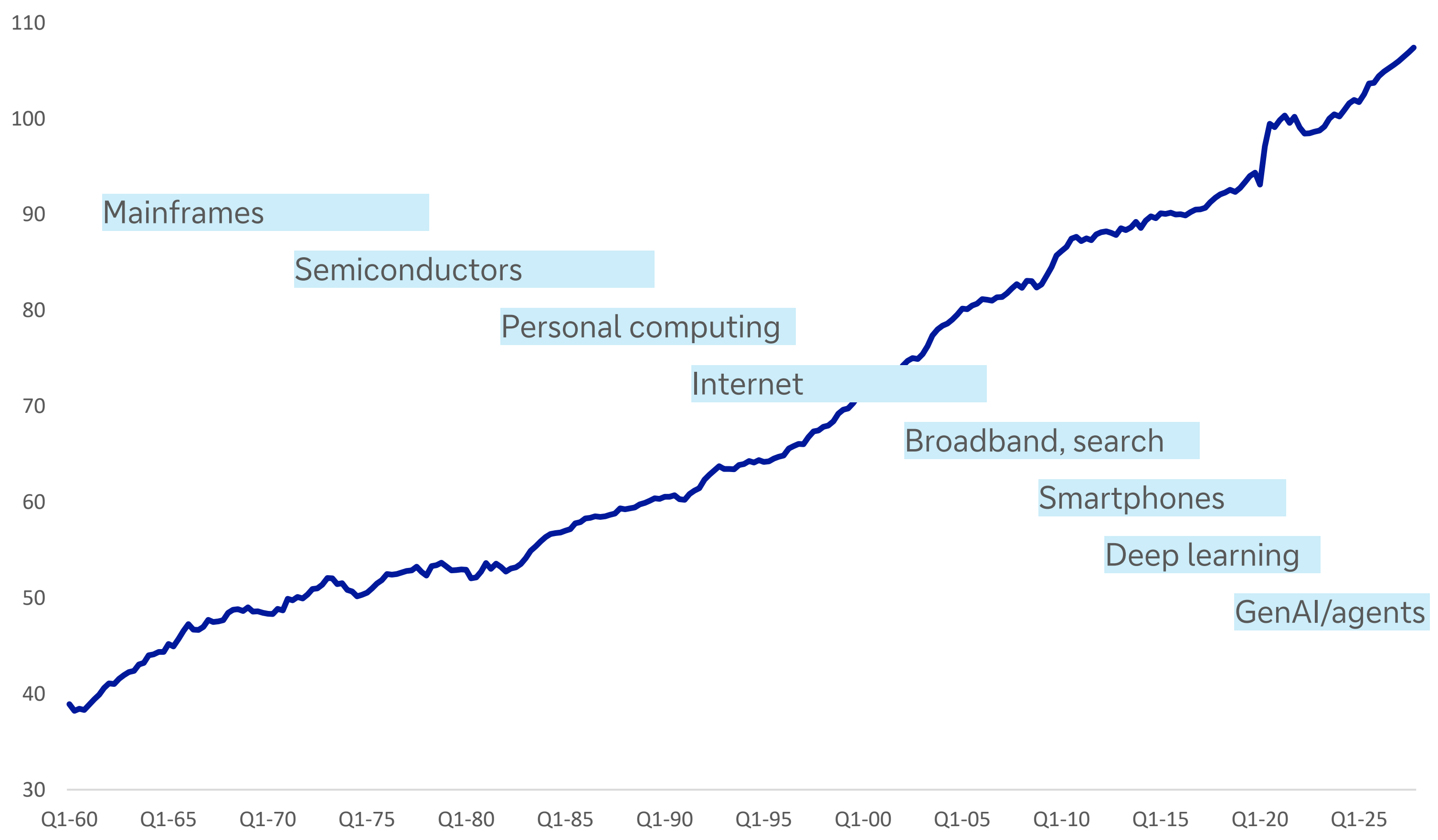




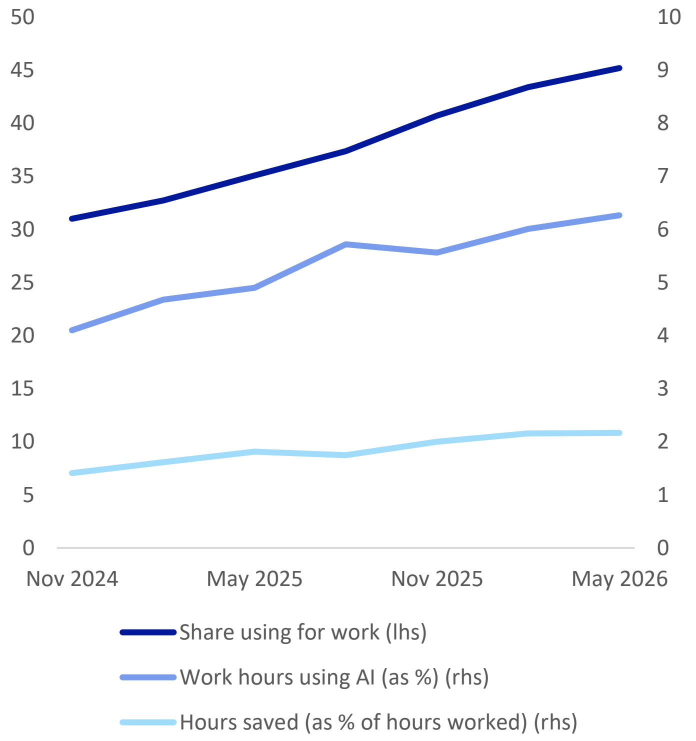
9. Tech revolutions take time to pay off, making them hard to see in long-term data

General purpose technologies have a productivity J-curve, generating costs before results

US productivity index since 1960 (2021 = 100)



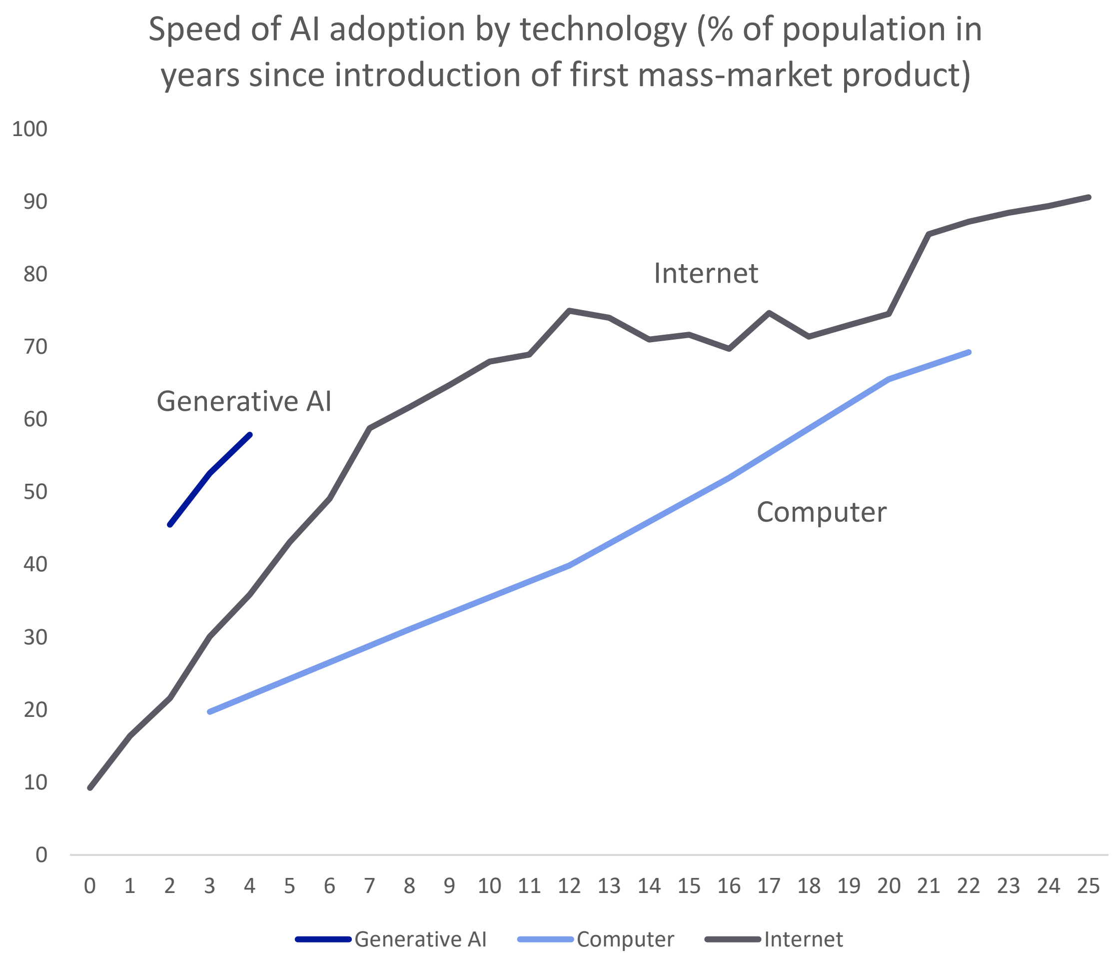
Fewer than half of US workers say they are using AI at work, with a mean of just 6% of time and saving 2% of hours

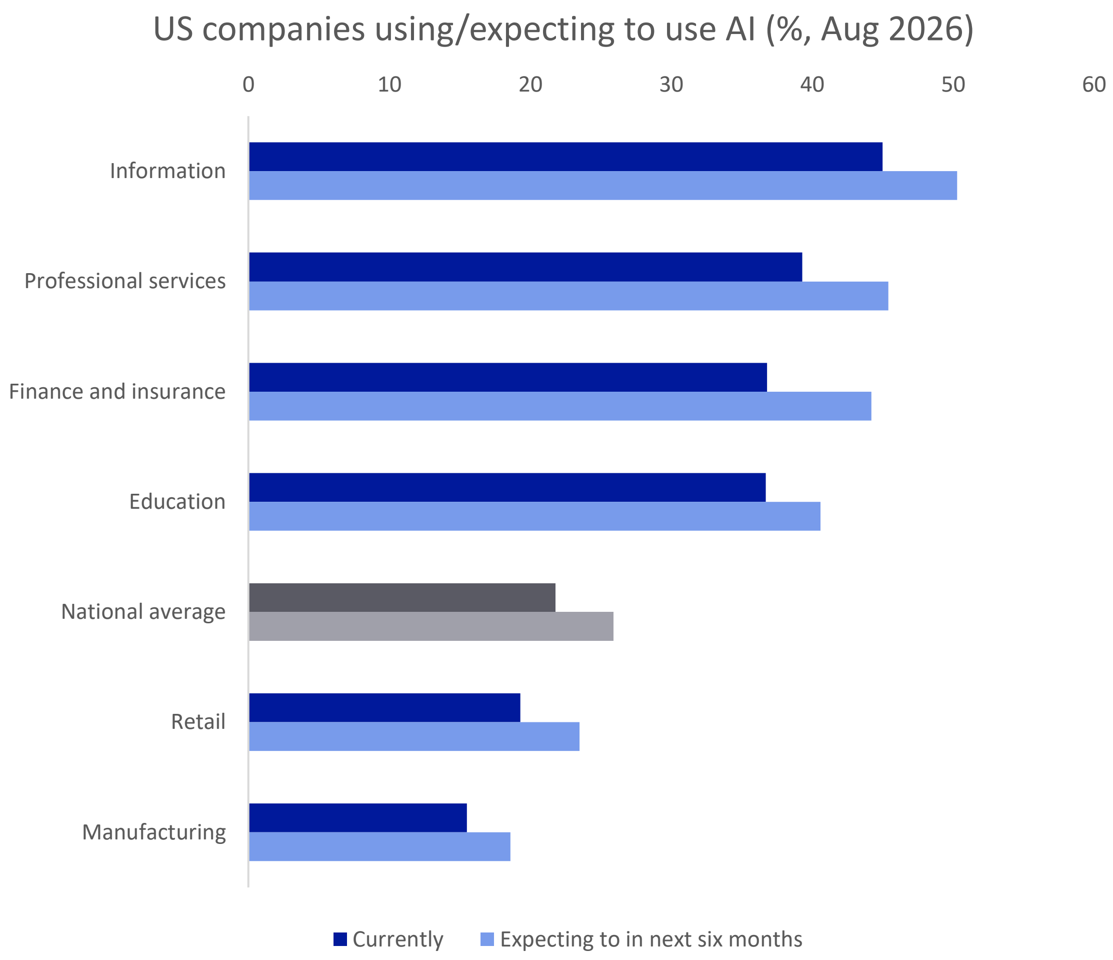


Source: OECD, GenAI Adoption Tracker, Deutsche Bank Research; blue bars in productivity chart give rough indication of notable tech eras

10. Consumers have adopted AI at a record pace but monetisable business use is slower

Downloading an app takes minutes; reorganising enterprises round new tech takes years

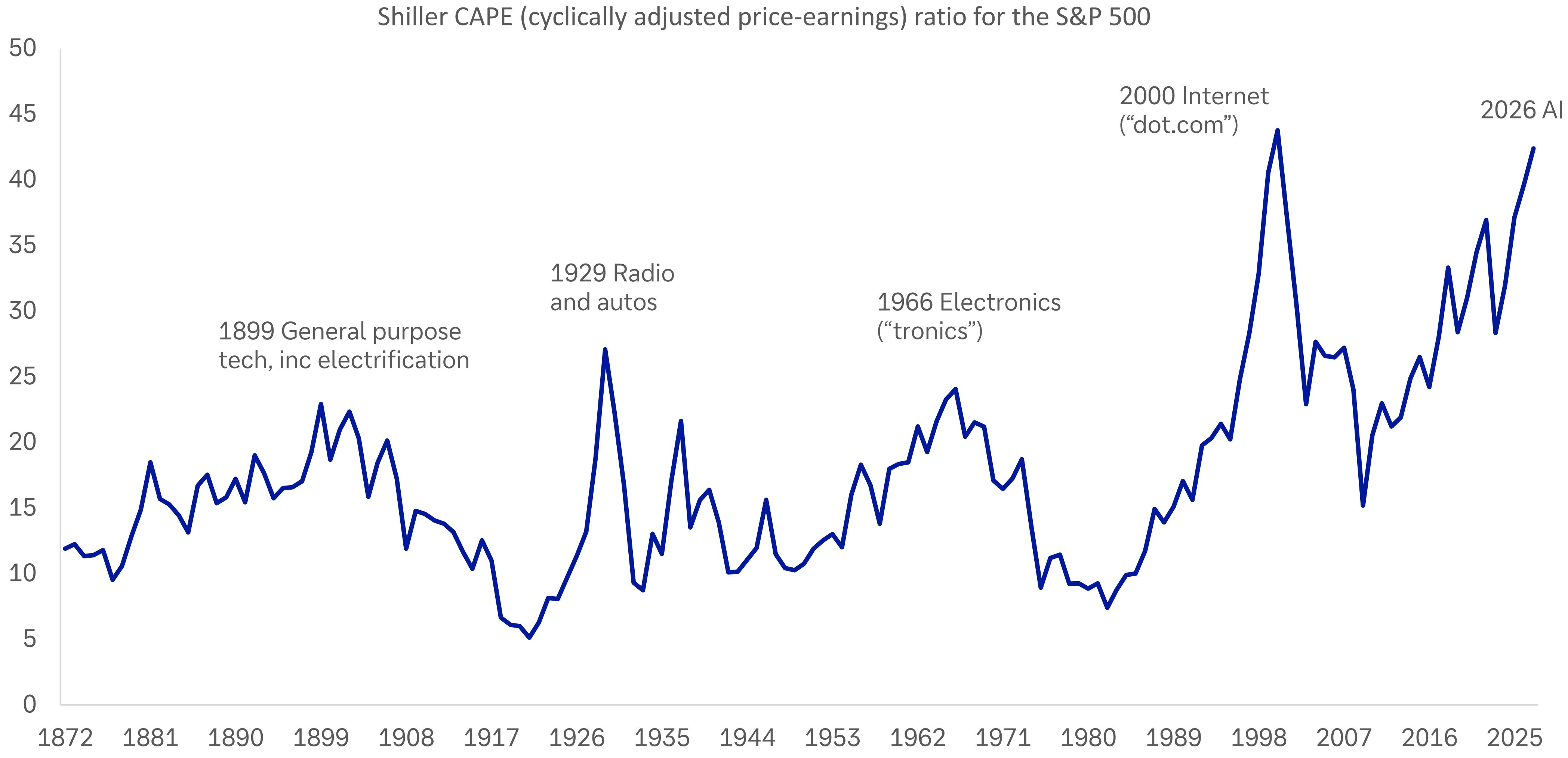




Source: GenAI Adoption Tracker, US Census Bureau, Deutsche Bank Research

11. The biggest valuation booms are often tech booms – like today

Peaks have mostly coincided with transformative technologies

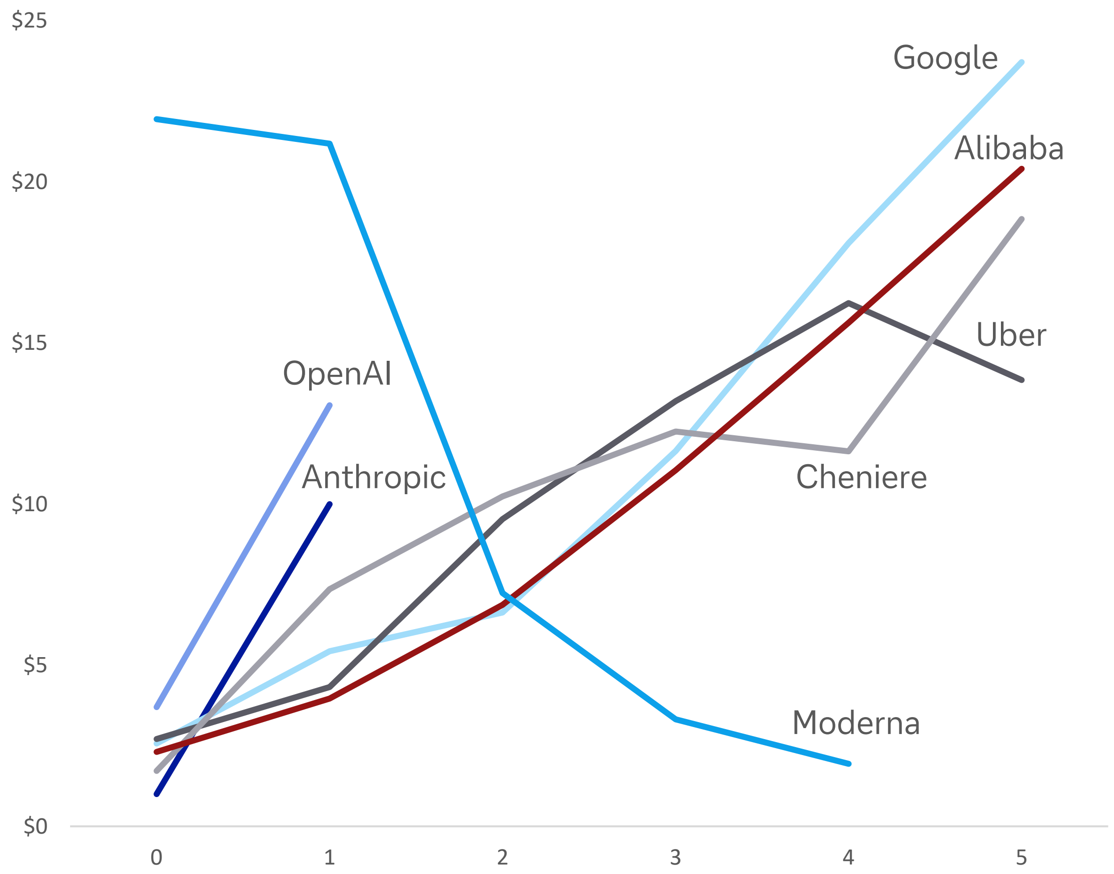


Source: Robert Shiller, Deutsche Bank Research

12. This time is different – but it will need to be even more different to meet projections

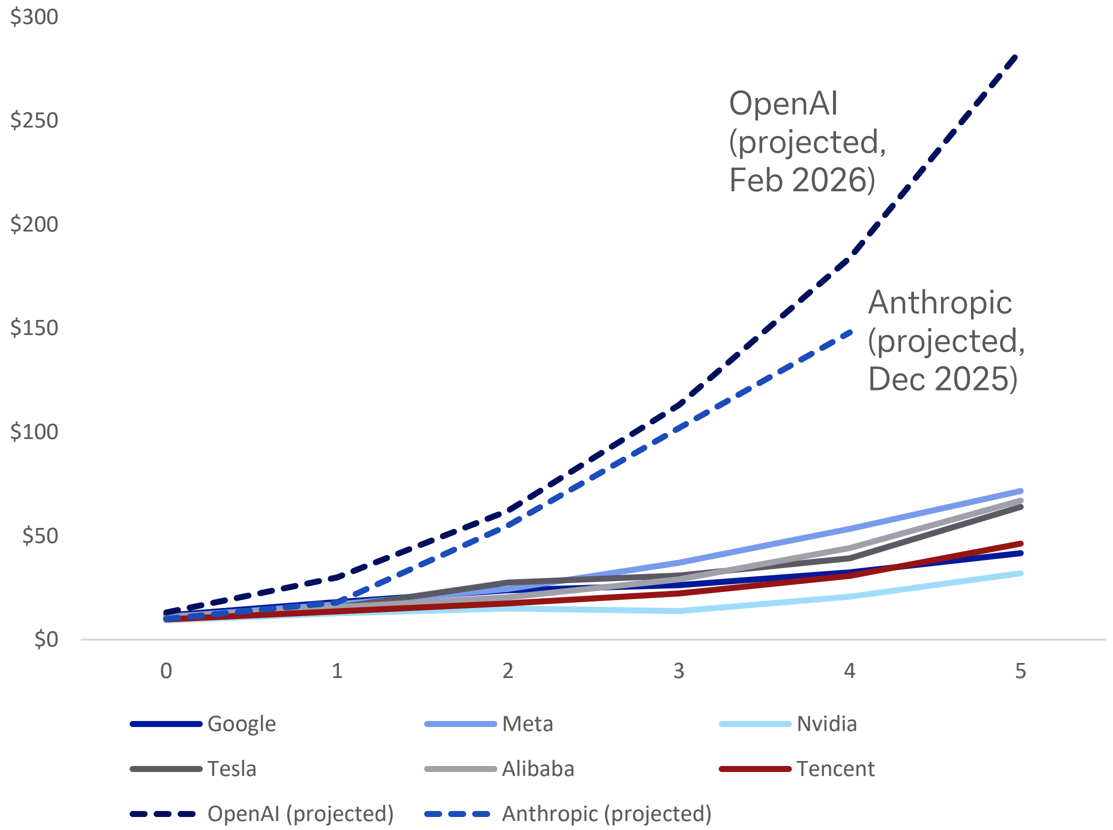
OpenAI and Anthropic are exceeding record revenue growth set by peers, at least for now

Annual revenue since exceeding $1bn milestone (USD bn)



Source: Bloomberg Finance LP, media reports, Deutsche Bank Research. Note: Year 0 shown as first year revenue exceeded $1bn (2025 USD). For OpenAI and Anthropic that was reportedly 2025. Moderna nominal revenue spiked from $800m in 2020 to $18.5bn in 2021, due to Covid vaccine. Cheniere revenue spiked with 2022 energy crisis.

Annual revenue since exceeding approx $10bn revenue, plus projected revenue for OpenAI/Anthropic (USD 2025)

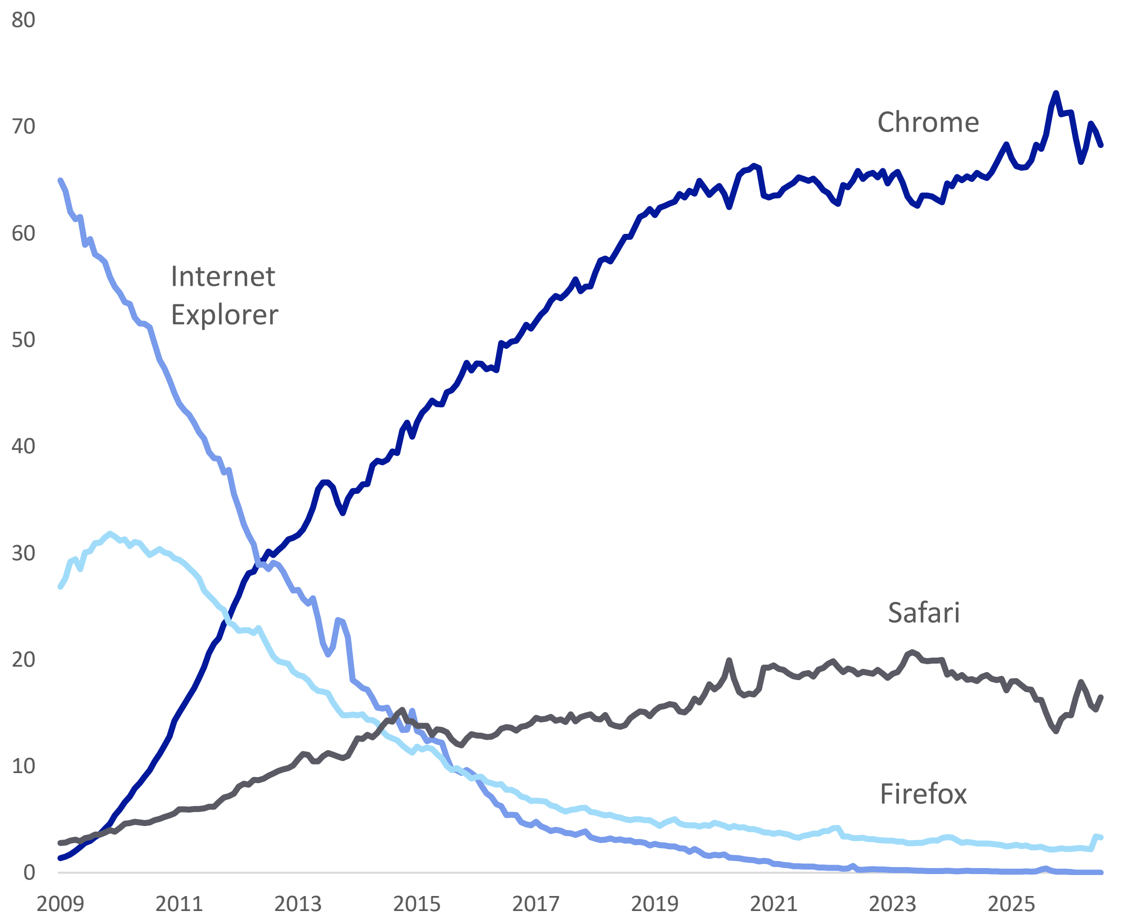


Source: Epoch, Bloomberg Finance LP, The Information (Feb 2026, citing recent updates to investors from OpenAI and Anthropic), Deutsche Bank Research. Note: Year 0 is first year revenue was approx $10bn (2025 USD)

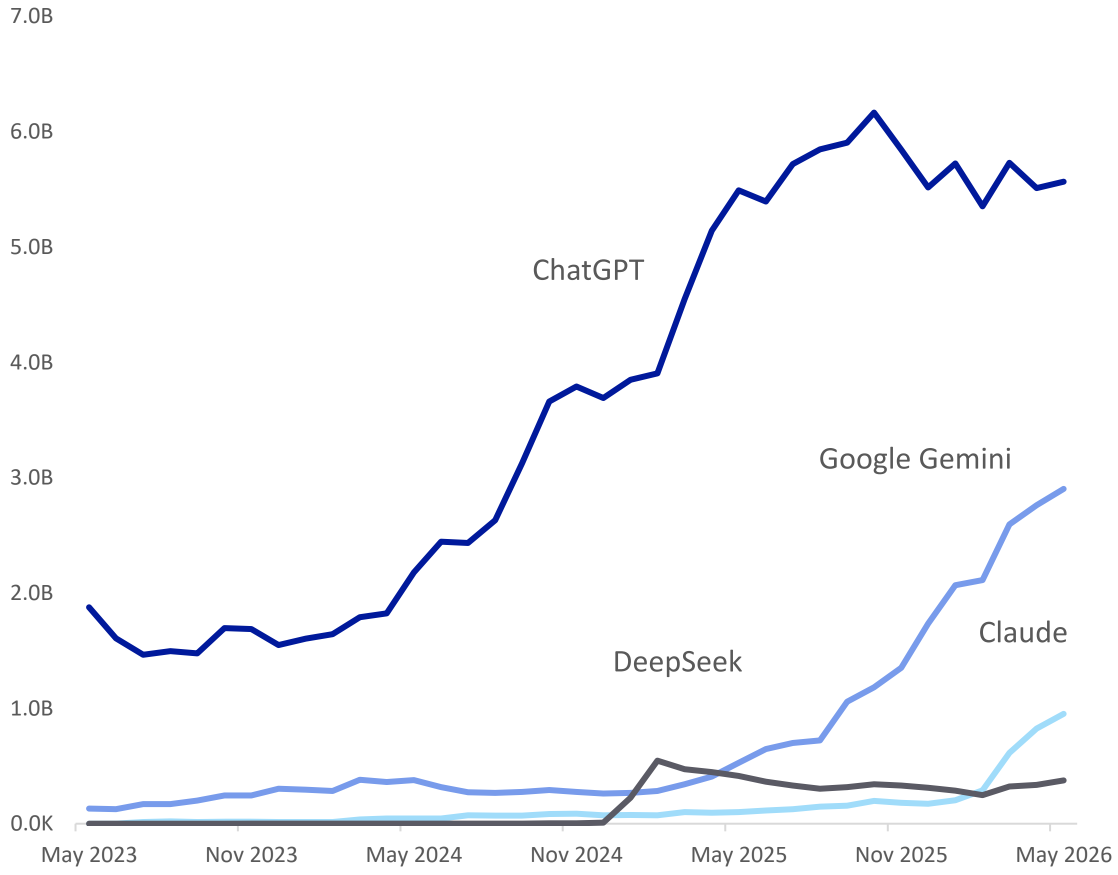
13. Early leads do not guarantee long-term winners

For example, top browser Internet Explorer overtook Netscape, then it too was overtaken

Browser market shares (%) (2009 to today)



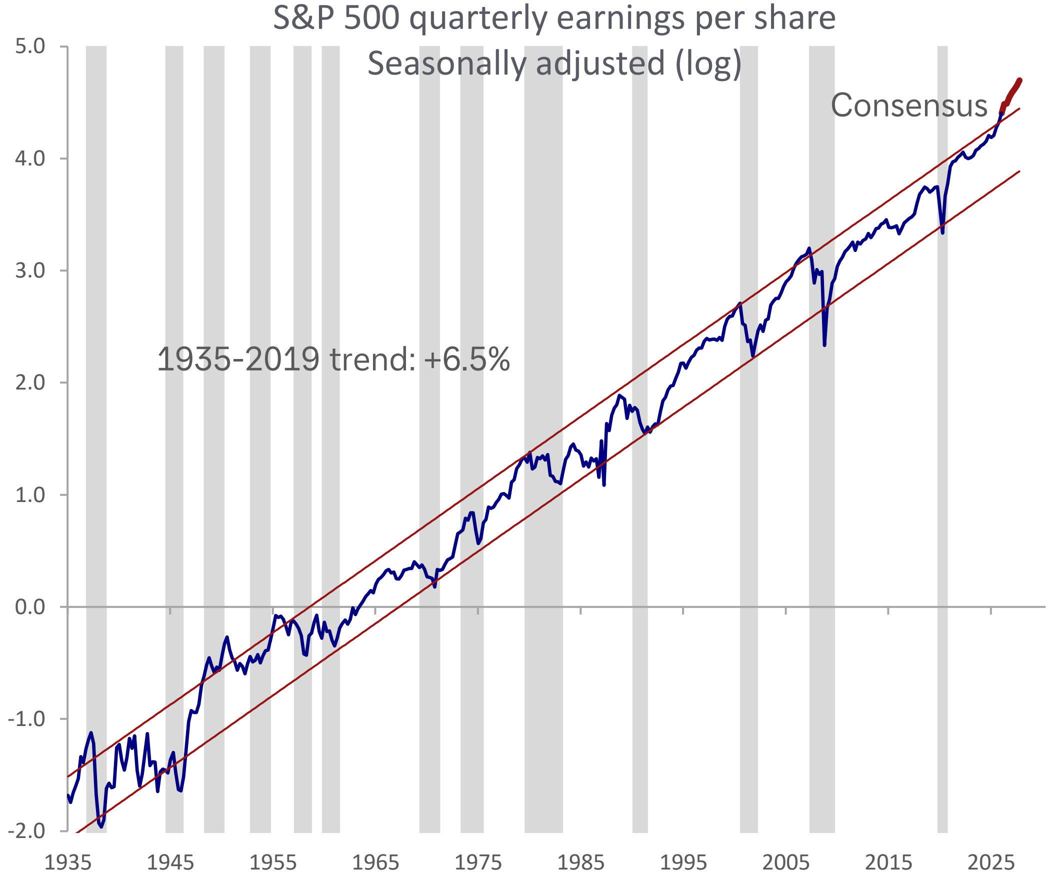
Top four generative AI platforms: monthly visits worldwide (desktop and mobile)

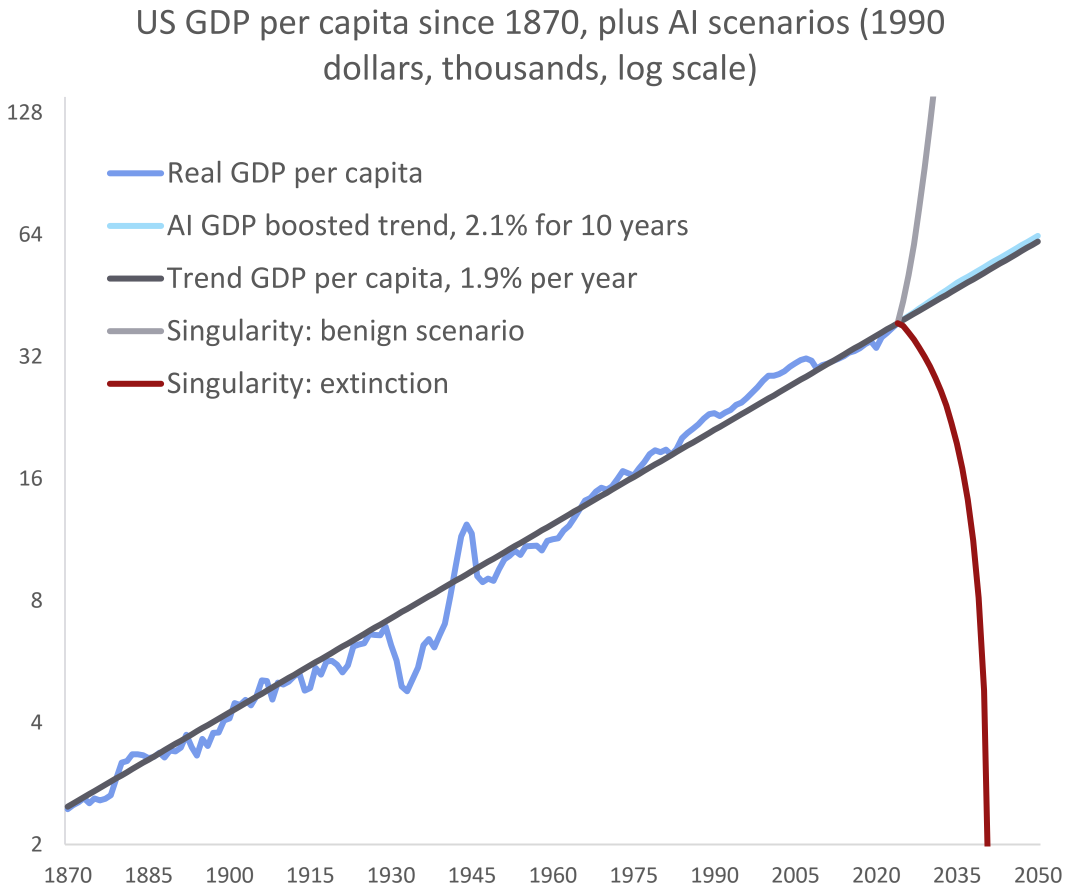


Source: Statcounter, Similarweb, Deutsche Bank Research

14. Long-term trends tend to persist through wars, recessions – and tech revolutions

Even so, some are considering more extreme scenarios this time… from utopia to extinction





Source: Bloomberg Finance LP, Federal Reserve Bank of Dallas, Deutsche Bank Research. Grey shaded bars in S&P quarterly earnings chart represent recessions

Appendix 1

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Deutsche Bank
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Deal or no deal, investors are both concerned and excited

August 11, 2026

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Executive summary

Five months into the US-Iran conflict and the strength of corporate earnings makes it seem like the optimists have won. But things are more complicated. This report analyses proprietary data from our dbDataInsights team and shows that the issue of investing against the backdrop of a volatile energy market is highly polarising for people. Indeed, of all the investment themes we track, energy has one of the highest rates of both optimism and pessimism.

So, we have an odd situation: On one hand, global corporate earnings are incredibly strong despite the energy shock this year. On the other hand, the Iran conflict is still a massive problem for the world economy. An amicable agreement thus remains elusive.

Three findings from our analysis stand out.

— Energy – whether measured through prices or transition – is now one of the most-cited investment themes for US investors behind AI, and is prompting portfolio changes on par with technology and geopolitics.

— Energy prices carry one of the highest perceived risks of any theme we tested among UK investors.

— Our megatrend model shows that energy disruption (as a medium-term trend) has remained a positive force for the world economy this year. That pattern closely echoes the 1970s oil shocks.

Energy shocks are unambiguously bad for consumers and growth in the short term. But over longer time frames, our model shows that energy disruption has tended to be a net positive for markets and economies because it accelerates diversification of supply and pushes economies to generate more GDP per unit of energy consumed. We see early evidence of a similar dynamic beginning today.

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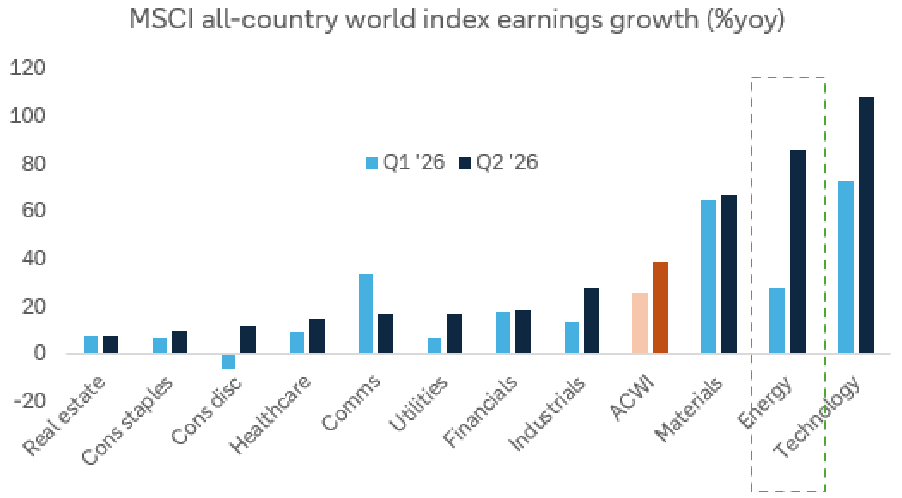
A serious issue ... but surprising resilience

The closure of the Strait of Hormuz has not been the economic catastrophe that many expected five months ago. In fact, corporates earnings have just had one of their best quarters outside of recession recoveries. For the companies in the MSCI all-country world index, median earnings growth is running at 16.5% yoy in Q2, well above the 14.7% in Q1. Energy has been a big part of that, with earnings growth, at a sector level, surging 86% in Q2, about triple the growth recorded in the prior quarter.

Figure 1: Earnings growth has been very strong in Q2



Source: Bloomberg Finance L.P., Deutsche Bank Asset Allocation

Economies have fared less well than companies, but they remain remarkably resilient given the harsh predictions that came from some quarters in early March. Of course, stockpile releases and other buffers have helped smooth the disruption – and as these wear down further, larger problems may arise.

So, now we have a curious juxtaposition of outlooks. On one hand, corporate earnings are proving resilient and the long-term focus on energy security has helped make many of the associated equities good investments. On the other hand, the Iran conflict is still a massive problem for the world economy.

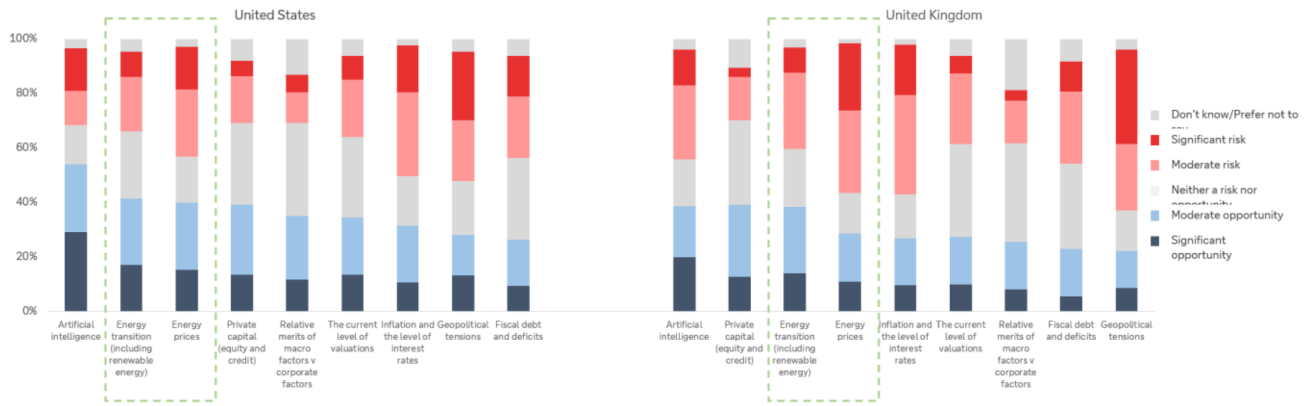
What investors think about the effect of energy on their portfolio

Our dbDataInsights survey on the topic was carried out in April and May of this year and polled a representative sample of people in the US and the UK. The responses indicate that energy has become one of the two dominant investment themes of 2026 – alongside, and closely intertwined with, artificial intelligence.

Energy is a high-conviction theme – in both directions

When we asked investors whether they see various investment themes as a risk or an opportunity for 2026, energy transition and energy prices ranked just behind artificial intelligence as the most-cited opportunities in both the US and the UK. The below chart shows a selection of the themes.

Figure 2: Survey question: "For each investment theme listed below, please indicate whether you perceive it as a risk or an opportunity for investors in 2026"



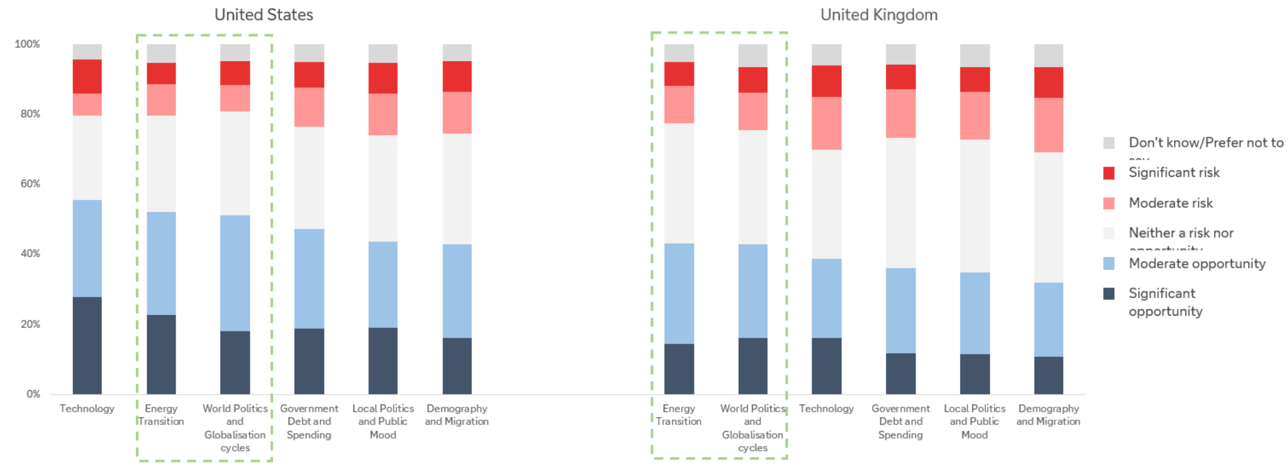
Source: dbDataInsights, Deutsche Bank

Energy is already changing portfolios

Conviction about energy appears to be translating into action – particularly when we think about it from a medium- or long-term point of view. To assess this, we fell back on our megatrend model that assesses the development of the six key megatrends that are driving the world's economies and markets.

When we asked investors how likely they are to change their portfolio over the next 12 months because of each megatrend, a dominant theme was energy and geopolitics as the below chart shows.

Figure 3: Survey question: "Within the next 12 months, how likely are you to make changes to your investment portfolio due to the following thematic megatrends?"

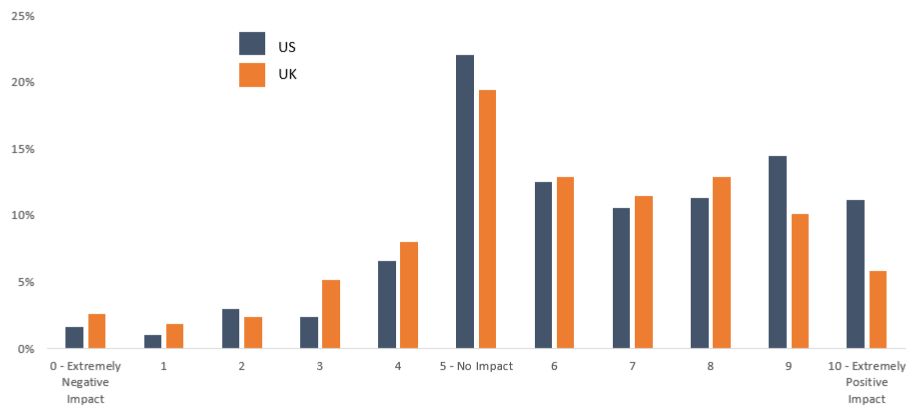


Source: dbDataInsights, Deutsche Bank

Taken together, the two survey questions tell a consistent story: energy has moved from a background, structural megatrend to one that investors are actively trading around, much as technology has been for the AI boom. The difference is that, unlike technology, energy's near-term path is being set as much by missile strikes and shipping insurance premia as by fundamentals.

Investors appear to sense this: when we asked how the energy transition will affect their main portfolio over the next three years, responses skewed firmly positive.

Figure 4: Investors' anticipated impact of the energy transition on their main investment portfolio over the next 3 years

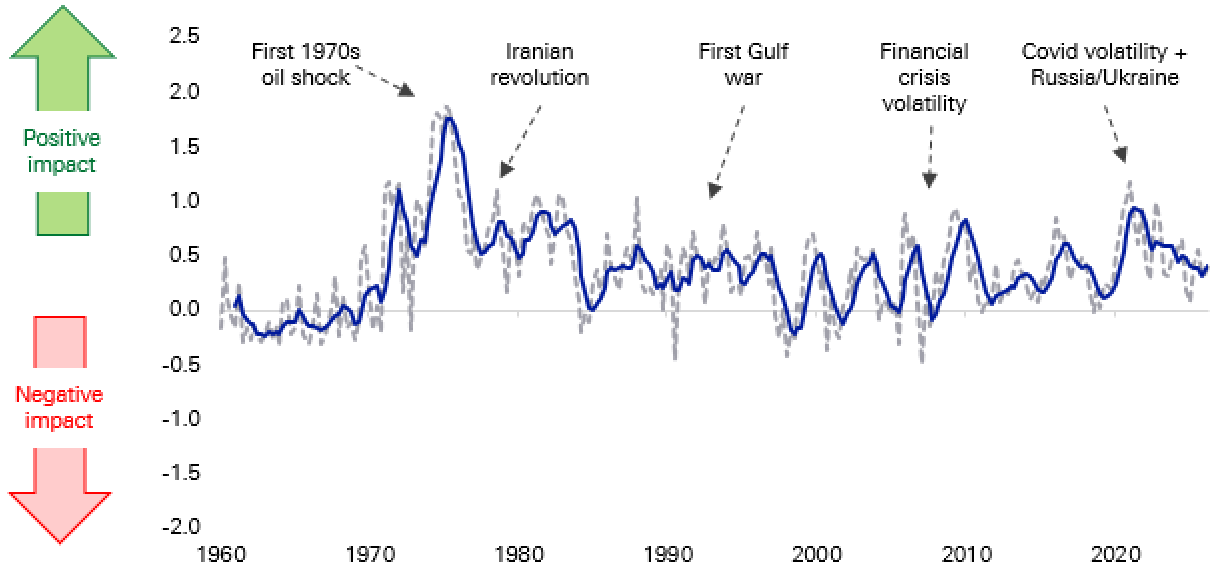


Source: dbDataInsights

Energy as a megatrend: why the medium-term signal is so positive

Energy is one of the six long-term trends that feeds into our megatrend model that uses almost 100 data series since 1957 to track their impact on economies and markets. Energy is a nuanced megatrend as, unlike several others, there is a big difference between the effect of short-term disruption and long-term gain. Indeed, prior periods of energy disruption have been incredibly good for the long-term energy trend, even if the short-term disruption was incredibly painful for consumers and economies.

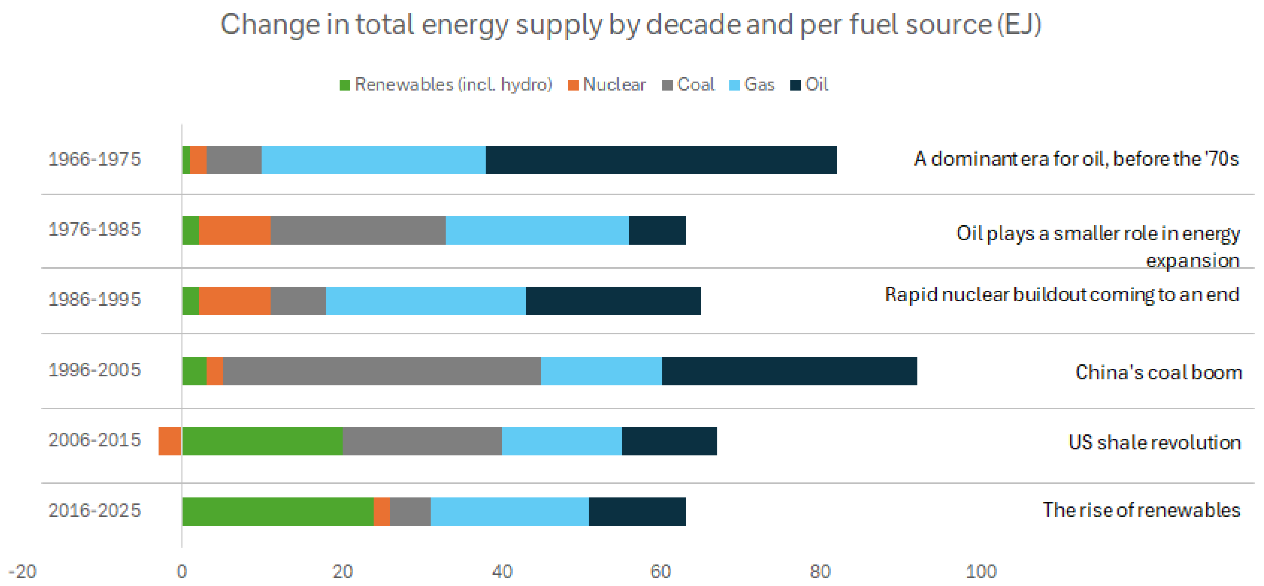
Figure 5: Our energy megatrend indicator has generally been positive as the world's economies diversify their supplies at the same time that innovation makes the economy more energy efficient



Source: Factset, Bloomberg Finance L.P., Fineon, EIA, Deutsche Bank Research

Viewed through a medium-term lens, the last five months of conflict in the Middle East can be viewed as likely being a long-term positive for economies and markets. That is because the Iran war is accelerating exactly the kind of diversification our model rewards: European and Asian buyers have moved further into US, Guyanese and West African crude and LNG since the Strait's closure; several G20 governments have fast-tracked strategic reserve releases and non-Gulf supply agreements; and renewed urgency has attached to the 'friend-sufficient' push for critical minerals and energy inputs that began during the US-China strategic rivalry.

Figure 6: Shocks over the decades have defined several eras for the evolution of the energy mix



Source: Energy Institute

The 1970s is something of a parallel

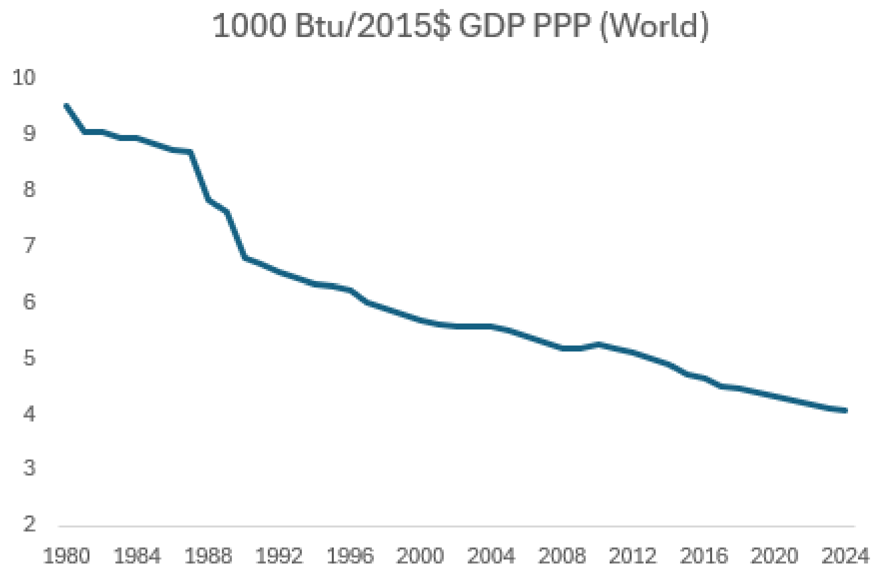
The closure of the Strait of Hormuz in 2026 and the disruption to Russian gas flows into Europe in 2022 both echo the twin oil shocks of 1973 and 1979, when OPEC's move to set its own prices followed by the Iranian revolution, sent crude to ten times its early-1970s level and permanently altered the trajectory of global oil demand.

Global oil consumption growth slowed from 7.8% before 1973 to 2.2% pa for most of the rest of the decade (on calculations from the Energy Institute). It outright declined after the second and took a decade for global oil consumption to recover to its 1979 level.

Whilst the 1970s shocks created havoc for both individuals and economies, they accelerated the pace at which the world changed how it generated economic output. The energy intensity of GDP fell 3.3% in 1980 (the largest single-year improvement in the Energy Institute's data series). That adjustment reflected both behavioural change and, over the following decade, structural investment in fuel efficiency and the phase-out of oil-fired power generation.

Since 1980, the energy intensity of GDP has roughly halved as shown in the following chart.

Figure 7: The energy intensity of GDP has been steadily falling – partly because of efficiency and innovation



Source: EIA

This trend is captured in our megatrend model which shows the medium-term payoff from decades of energy disruption. Those forces continue today. Every episode of chokepoint risk, from Ukraine in 2022 to Hormuz in 2026, adds another increment to the diversification of global energy supply and another incentive for energy innovation across economies.

Conclusion

Five months into the US-Iran conflict war, investors are highly polarised about the investment merits of investing against the backdrop of energy volatility. However, those that are positive are very positive. Our megatrend model suggests that reaction is rational: like the 1970s oil shocks, the medium-term legacy of this conflict is likely to be faster energy diversification and a further, durable improvement in the energy intensity of GDP, even though the near-term path for prices is likely to stay volatile.

As a result, energy security and supply diversification should keep attracting capital regardless of where the oil price settles. The beneficiaries are likely to cascade up and down the value chain including, LNG infrastructure, non-Gulf production, renewables, strategic reserves and grid resilience as countries and companies all try to diversify their chokepoint risk.

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Korea: From Industrialization to Internationalization

August 10, 2026

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Executive Summary

Korea sits at the intersection of some of the most important structural forces reshaping the global economy. Re-industrialization, AI infrastructure investment and supply-chain diversification are directing capital toward industries where Korea already possesses scale, technological leadership and established industrial ecosystems. At the same time, policymakers are reinforcing these advantages by supporting strategic industries and expanding Korea's industrial footprint globally.

Yet a paradox remains. Korea's economic influence has expanded much faster than its financial influence. Despite accounting for roughly 2% of global GDP and playing a central role in several strategic industries, the Won remains only a modestly used international currency and Korean assets continue to trade at valuation discounts. Korea's internationalization agenda seeks to narrow this gap, building a financial system that better reflects the country's growing importance in global trade, manufacturing and technology.

In this report, we argue that:

- **The world is increasingly investing in industries where Korea already leads,** as re-industrialization, AI investment and supply-chain diversification redirect capital toward Korea's areas of strength.
- **Korea already possesses what others are trying to build,** with its strengths in manufacturing, technology, innovation and competitiveness.
- **Internationalization is the next stage of development,** helping Korea retain and intermediate a larger share of the wealth it creates.
- **Korea is pursuing a distinct and phased model of internationalization,** combining market-access reforms, greater offshore usability and gradual liberalization while preserving financial stability.
- **The key market implication is a gradual narrowing of the gap between Korea's economic footprint and financial footprint,** as stronger industrial performance increasingly translates into deeper capital markets, a more internationally used currency and stronger demand for Korean assets.

The world is investing in the industries where Korea already leads

Industrial policy has returned to the centre of global economic strategy. Across the US, Europe, Japan, China and much of Asia, governments are deploying subsidies, tax incentives, localisation requirements and strategic investment programmes to strengthen domestic manufacturing, secure access to critical technologies and reduce supply-chain vulnerabilities. While the policies differ across jurisdictions, the direction of travel is increasingly aligned - resilience, technological leadership and economic security now rank alongside efficiency as core policy objectives.

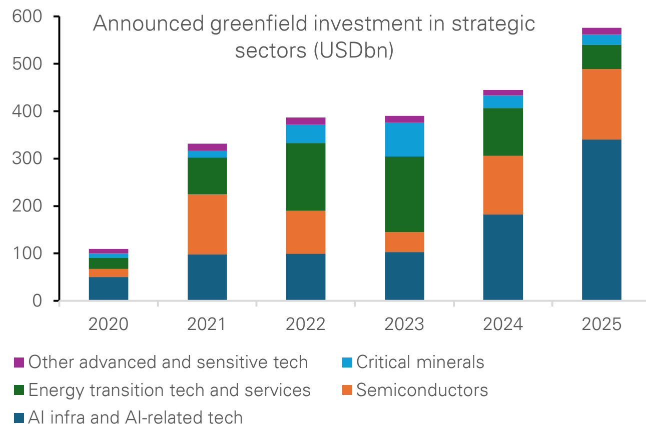
Figure 1: Industrial policy comparison — At-a-Glance

	Policy direction	Strategic sectors	Main objective	Incentive model
Advanced economies				
US	Reshore critical technology production	Chips; EVs/batteries; clean energy; defence; critical minerals	Secure strategic supply chains with trusted partners	Subsidies, tax credits, grants, loans, export controls
EU	Make decarbonisation an industrial growth engine	Net-zero industry; clean tech; heavy industry; raw materials	Build competitive clean industry while protecting competitiveness	State-aid flexibility, EU funding, carbon rules, faster permits
Japan	Strengthen economic security and low-carbon industry	Chips; batteries/EVs; defence; aerospace; nuclear/clean energy	Secure advanced inputs and reinforce defence/energy resilience	Targeted subsidies, tax breaks, procurement, energy investment
North Asia				
China	Pursue self-sufficiency and technology leadership	Chips; EVs/batteries; shipbuilding; green tech; critical minerals	Reduce external vulnerability while preserving export strength	Subsidies, cheap credit, tax relief, procurement, export controls
Korea	Scale national champions in strategic technologies	Memory chips; AI; EV batteries; autos; shipbuilding; defence; nuclear	Hold top-tier chip position and diversify growth engines	Mega capex funds, policy credit, tax credits, R&D and talent programmes
Taiwan	Extend semiconductor leadership into AI hardware	IC design/manufacturing; AI servers; smart manufacturing; digital sectors	Use AI + chips to deepen the innovation ecosystem	R&D credits, shared labs, pilot lines, talent/startup support
South Asia				
India	Scale manufacturing and domestic value-add	PLI sectors; electronics; autos/EVs; chips; pharma; capital goods	Become a high-value manufacturing and export hub	Output-linked incentives, sector schemes, infrastructure/logistics funding
Indonesia	Move nickel downstream into EV value chains	Nickel; battery materials; cells; EV assembly; clean-energy metals	Shift from raw materials to integrated battery/EV production	Export bans, smelter controls, industrial parks, state-firm ecosystems
Malaysia	Upgrade from assembly to higher-value manufacturing	E&E; semiconductor back-end; advanced materials; aerospace; pharma	Move up technology and Industry 4.0 capability	Tax incentives, grants, roadmaps, cluster facilitation
Philippines	Revive manufacturing around electronics and components	Electronics packaging; autos; aerospace components; broader manufacturing	Deepen regional supply-chain participation beyond services	SEZ incentives, FDI promotion, sector roadmaps, streamlined approvals
Singapore	Anchor high-value advanced manufacturing	Precision engineering; chips; pharma; robotics; automation; digital manufacturing	Keep high-value production linked to R&D and HQ functions	Grants, co-funding, R&D incentives, skills schemes, bespoke EDB packages
Thailand	Use EEC and S-curves to upgrade industry	EVs; smart electronics; aviation/logistics; robotics; bio/medical industries	Become a regional hub for upgraded manufacturing	SEZ tax breaks, infrastructure, premium sector incentives
Vietnam	Build supporting industries and semiconductor capability	Electronics support; mechanical/autos; textiles/footwear; high-tech; chips	Raise localisation and integrate deeper into global supply chains	Cost-sharing subsidies for R&D, technology and workforce spending

Source : Deutsche Bank Research

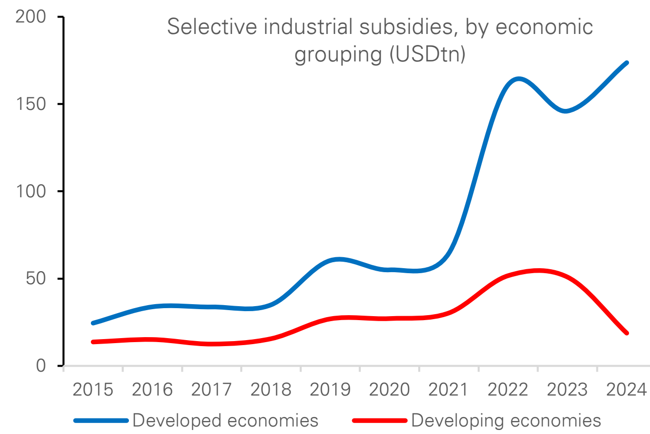
The scale of this shift is already visible in global investment flows. According to UNCTAD, strategic sectors—including semiconductors, clean energy, advanced manufacturing, defence and digital infrastructure—accounted for 44% of global greenfield investment in 2025, up from just 16% in 2020. Over the same period, the value of announced projects rose from $109bn to $576bn, highlighting the growing concentration of capital in strategic industries. This is creating one of the largest industrial investment cycles in decades and directing investment towards precisely the sectors where Korea has already established globally competitive capabilities.

Figure 2: Growing investments in strategic sector globally



Source : Deutsche Bank Research, UN Trade and Development (UNCTAD). Note: Data for 2025 are annualized on the basis of information available as of 30 November.

Figure 3: Increase in the use of subsidies to support growth in the strategic sector



Source : Deutsche Bank Research, UN Trade and Development (UNCTAD), based on New Industrial Policy Observatory (NIPO). Note: Only subsidies worth more than $10mn are recorded in the NIPO database.

Three structural forces are now converging in Korea's favour.

- **Reshoring, ally-shoring and rearmament are increasing global demand for industries in which Korea already holds a competitive advantage.** Semiconductors, shipbuilding, defence equipment, batteries and critical minerals have become central to national-security and supply-chain strategies. Post-Ukraine European rearmament has increased demand for reliable defence suppliers, while the US and its allies are seeking trusted partners to rebuild strategic supply chains. Korea already possesses many of the industrial capabilities that advanced economies are now attempting to reshore or source from aligned partners.
- **The global AI capex supercycle is providing a major terms-of-trade uplift**. Hyperscaler capital expenditure is estimated to be growing at close to 70% YoY through 2024-26, creating strong demand for GPUs, HBM, DRAM and related semiconductor infrastructure. As large language models become increasingly complex and agentic AI requires greater processing power and speed, demand for advanced memory chips is rising sharply. Samsung Electronics and SK Hynix place Korea at the centre of this cycle, positioning the country as one of the most direct beneficiaries of the global AI infrastructure buildout.
- **China de-risking is reshaping supply chains in Korea's favour.** Western buyers are actively reducing reliance on Chinese supply chains across selected strategic sectors, including batteries, battery materials, vessels, robotics and critical minerals. Korea is one of the few economies with both the industrial scale and geopolitical alignment to absorb some of the production, investment and market share being redirected away from China. While Chinese overcapacity remains a competitive challenge in some industries, the global effort to diversify away from China is simultaneously creating opportunities in strategic sectors where technology, reliability and security considerations matter more than cost alone.

This convergence matters. Korea is not benefiting from one isolated export cycle. **Rather, it sits at the intersection of ally-shoring, AI infrastructure investment and China supply-chain diversification—three structural forces that are likely to persist for years.** In many respects, the world is increasingly investing in precisely the industries where Korea already possesses scale, technological leadership and established industrial ecosystems.

Figure 4: Korea ranks among the world's best positioned economies for re-industrialisation

Rankings (1 = Best)													
Metric	US	EU	Japan	China	Korea	Taiwan	Singapore	Malaysia	Indonesia	Thailand	Vietnam	Philippines	India
Manufacturing value added (% of GDP)	13	11	8	3	2	1	9	6	7	5	4	10	12
Share of high-tech exports	11	10	7	8	2	1	3	6	13	9	5	4	12
Current account balance (% of GDP)	13	8	5	6	4	1	2	9	10	7	3	12	11
Government debt (% of GDP)	12	10	13	11	5	2		8	4	7	3	6	9
Gross capital formation (% of GDP)	11	13	6	1	5	7	10	9	3	12	4	8	2
Total surplus capacity (MW)	3	2	5	1	6	10	13	12	7	9	8	11	4
Electricity price (USD/kWh)	8	12	11	4	5	10	13	7	1	6	2	9	3
Business infrastructure	1	7	6	3	5	4	2	8	12	9	10	13	11
Global talent competitiveness index	3	4	5	8	6	2	1	7	12	11	10	9	13
Regulatory quality	2	5	3	11	6	4	1	7	8	9	13	10	12
Trade freedom	12	6	8	9	10	2	1	3	7	11	5	3	13
Broad competitiveness	3	10	9	4	6	2	1	5	13	7	8	12	11
Economic Complexity	6	7	1	5	3	2	4	8	13	9	12	11	10

Source :Deutsche Bank Research, CEIC, Haver Analytics, Insead, World Bank, The Heritage Foundation, IMD.org, Harvard Business School

Policy is also reinforcing these advantages. Rather than creating entirely new industries, Korean policymakers are largely focused on scaling existing strengths, securing critical inputs and supporting the global expansion of Korean industrial ecosystems. Recent initiatives have directed capital toward semiconductors, AI infrastructure, batteries, defence and other strategic sectors through policy financing, investment programs and regulatory support. At the same time, Korea is strengthening access to critical minerals, expanding overseas industrial partnerships and facilitating outward investment by Korean firms. The objective is not simply to support national champions, but to reinforce the broader ecosystem around them and ensure Korea remains embedded in the next phase of global industrial development.

Korea already possesses what others are trying to build

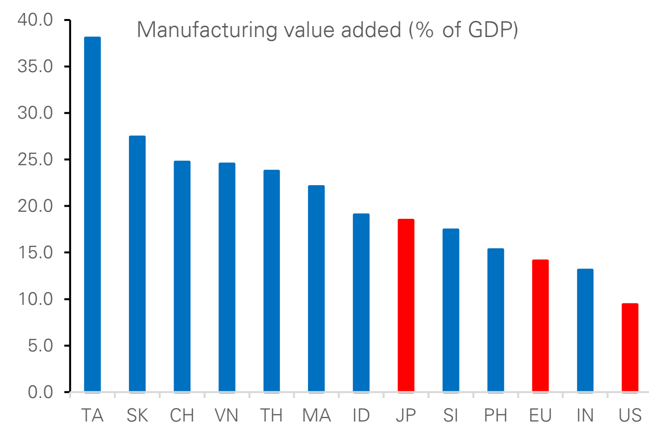
Few economies are better positioned to benefit from the new industrial order than Korea. Unlike many countries attempting to rebuild manufacturing capacity from a relatively low base, Korea already has one of the world's most advanced industrial ecosystems.

Few economies are better positioned to benefit from the new industrial order than Korea. Unlike many countries attempting to rebuild manufacturing capacity from a relatively low base, Korea already has one of the world's most advanced industrial ecosystems.

- **Industrial ecosystems matter.** Korea has built globally competitive positions across semiconductors, batteries, shipbuilding, defence, robotics, nuclear energy and advanced manufacturing. These industries are supported by a deep industrial ecosystem spanning engineering talent, R&D capabilities, specialised supplier networks and world-class infrastructure and relatively healthy public and external balances. These capabilities are difficult to replicate quickly and increasingly valuable in a world prioritising resilient and secure supply chains.

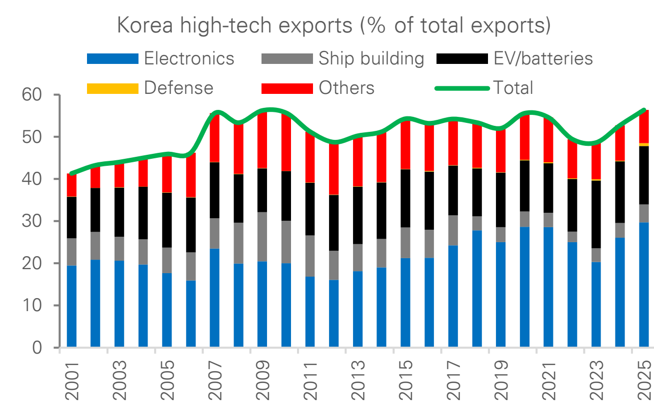
- **Starting from strength matters.** Manufacturing accounts for approximately 24.3% of GDP, among the highest shares in the OECD and roughly two times the level of the advanced economies. At a time when governments are seeking to expand domestic production of strategic goods, Korea enters the re-industrialisation cycle with an existing manufacturing base rather than having to build one from scratch.
- **Technology leadership matters.** Approximately 58% of Korea's exports are high-tech products, one of the highest shares among major economies, while semiconductors account for roughly 50% of high-tech exports. Korea also accounts for ~60% of global memory semiconductor production and 90% of global HBM supply, making it one of the most important beneficiaries of the AI infrastructure cycle.

Figure 5: Manufacturing accounts for approximately 24.3% of GDP, among one of the highest shares in the OECD and roughly two times the level of the advanced economies.



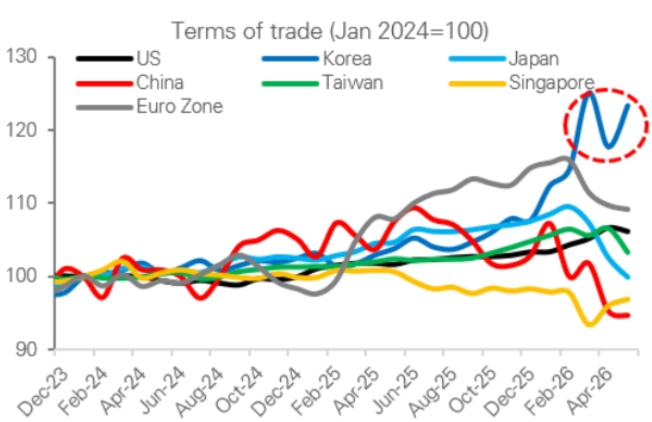
Source : Deutsche Bank Research, Haver Analytics

Figure 6: Korea's export engine increasingly driven by High-Tech industries



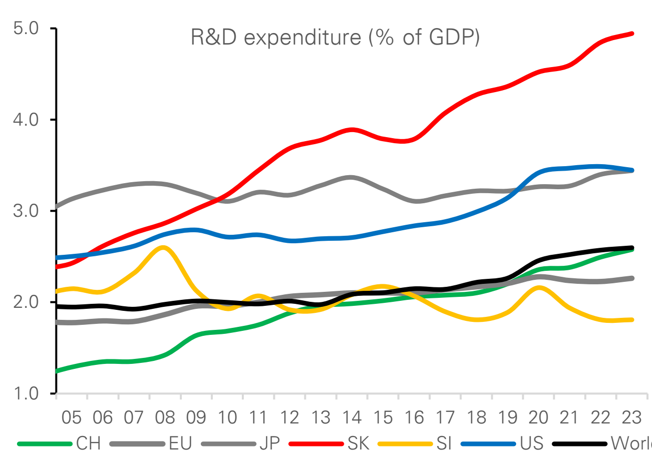
Source :Deutsche Bank Research, ITC

Figure 7: Significant improvement in terms of trade since 2024 for Korea



Source : Deutsche Bank Research, Bloomberg Finance LP, Haver Analytics

Figure 8: Korea has one of the highest R&D expenditures in the world



Source : Deutsche Bank Research, World Bank

- **Strategic industries matter**. Beyond semiconductors, Korea has built globally competitive positions in EV batteries, defence, advanced shipbuilding, robotics and nuclear energy. Korean firms account for ~30% of global EV battery capacity and 90% of global LNG carrier orders, while defence exports are increasingly benefiting from rising global security spending and rearmament efforts.
- **Energy increasingly matters.** Under South Korea's latest electricity plan, nuclear power is expected to account for approximately 35% of electricity generation by 2038, while carbon-free sources are projected to supply roughly 70% of total generation. The plan is designed in part to accommodate rising electricity demand from semiconductors, AI data centres and broader industrial electrification. As manufacturing and digital infrastructure become increasingly energy-intensive, access to reliable, scalable and lower-carbon power is emerging as an important component of industrial competitiveness.

Korea's advantages extend beyond individual industries. Through leadership in semiconductors, batteries, shipbuilding, defence and advanced manufacturing, Korea is becoming an increasingly important industrial partner for both advanced and emerging economies. As supply chains become more closely linked to economic security, reinforcing its role within the evolving architecture of the global economy. Rather than acting solely as an exporter of goods, Korea is increasingly participating in the development and expansion of industrial ecosystems across multiple regions through investment, technology transfer and supply-chain partnerships. In that sense, **Korea's industrial strengths are not only supporting growth and competitiveness at home but are also reinforcing its role within the evolving architecture of the global economy.**

Korea's industrial position remains strong, but its advantages should not be taken for granted. Despite leadership in semiconductors, shipbuilding, defence and advanced manufacturing, Korea remains dependent on imported energy, critical minerals and raw materials, leaving parts of its industrial ecosystem exposed to commodity and supply-chain shocks. Competition is also intensifying. China continues to move up the value chain across batteries, shipbuilding, robotics and segments of the semiconductor supply chain, while governments globally are deploying increasingly aggressive industrial policies to build domestic capabilities. Industrial leadership is rarely permanent. **Korea's re-industrialisation story therefore depends not only on existing strengths, but also on its ability to sustain innovation, secure critical inputs and preserve its technological edge over the coming decade.**

Internationalization as the next stage of development

Korea scores strongly on measures of industrial depth, export sophistication and external strength. Manufacturing value added, high-tech exports and current-account balances compare favourably with most regional peers. Yet Korea remains among the cheapest major Asian markets on P/E and P/B metrics, while the Won screens as undervalued on our valuation framework.

Figure 9: Korea's valuation paradox: strong fundamentals, cheap assets

	Economic Matrix			Financial Market Matrix				
	Manufacturing value added (% of GDP)	Share of high-tech exports (%)	Current account bal (% of GDP)	12M forward P/E ratio	ROE (%)	12M forward P/B ratio	Dividend yield	FX valuation
US	9	28	-4	21.3	20.0	4.6	1.2	5.7
EU	14	31	2	15.8	12.6	2.2	3.1	-1.6
Japan	18	40	5	22.4	12.0	2.7	1.5	-24.7
China	25	35	4	14.9	9.9	1.6	2.6	-5.3
Korea	27	59	7	6.9	10.5	1.5	1.9	-15.0
Taiwan	38	78	17	20.3	13.9	4.0	2.1	-5.3
Singapore	17	55	17	17.2	9.9	1.8	4.1	4.9
Malaysia	22	48	2	15.1	10.6	1.5	4.3	1.8
Indonesia	19	9	0	10.0	11.8	1.3	6.3	-7.8
Thailand	24	31	3	16.1	8.6	1.6	3.6	7.5
Vietnam	25	48	7	12.0	15.1	1.7	1.9	
Philippines	15	53	-3	9.9	11.6	1.2	3.7	2.9
India	13	23	-1	19.9	14.2	2.6	1.8	-6.6

Note: FX valuation is based on the simple average of three valuation models: FEER, DBEER, and PPP. Positive (negative) values indicate overvaluation (undervaluation).

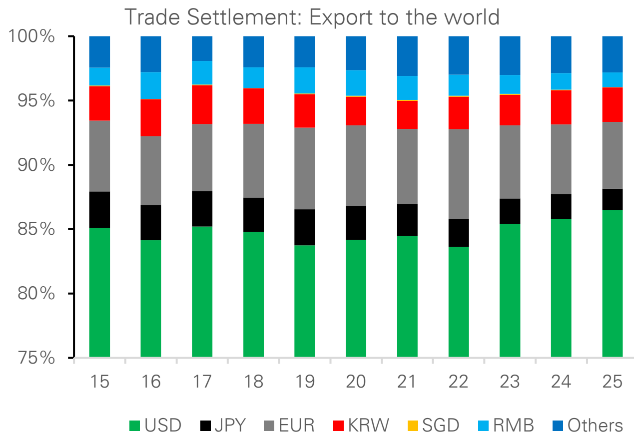
Source :Deutsche Bank Research, CEIC, Haver Analystics, Bloomberg Finance LP. Note: FX valuation is based on the simple average of three valuation models: FEER, DBEER, and PPP. Positive (negative) values indicate overvaluation (undervaluation).

Our preferred valuation framework suggests the Won remains materially undervalued relative to fundamentals. Korea's current account surplus could reach ~10.3% of GDP this year and remain above 5% over the medium term, well above its long-run average of around 3.5% of GDP. Under a FEER framework, which anchors exchange rates to external-balance sustainability, such an external position is inconsistent with current KRW levels and **implies that the Won is undervalued by ~ 10% in trade-weighted terms.**

More importantly, this result may prove conservative. Korea's current account improvement increasingly reflects a positive terms-of-trade shock driven by AI-related semiconductor demand, rising export sophistication and stronger pricing power. As a result, Korea's industrial renaissance may not only justify larger external surpluses, but also a higher equilibrium valuation for the Won.

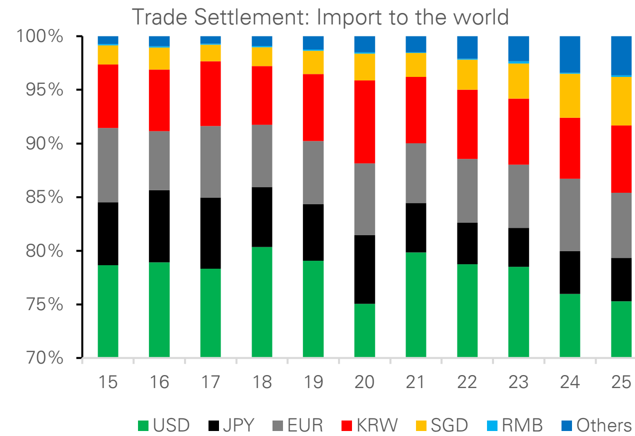
Besides asset valuation discounts, Korea's growing economic importance is not reflected in its financial system and currency yet. Today, Korea accounts for roughly 2% of global GDP, ranks among the world's largest exporters of advanced manufactured goods and is home to several globally strategic industries. Won accounts for approximately 0.9% of global FX turnover, which puts it at around 12th rank; while its role in international payments, funding markets and reserve holdings remains considerably smaller than Korea's relative economic quantum.

Figure 10: Korea's exports are still overwhelmingly settled in foreign currencies



Source : Deutsche Bank Research, Haver Analytics

Figure 11: The won plays only a limited role in Korea's import settlement

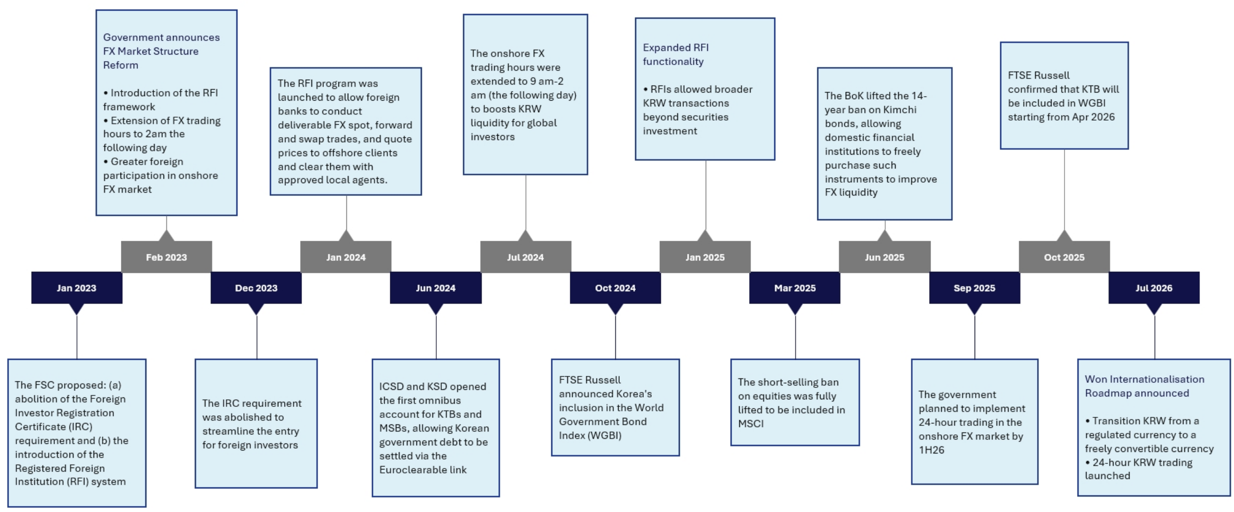


Source : Deutsche Bank Research, Haver Analytics

This gap between economic importance and financial influence helps explain the government's newly announced roadmap for Won internationalization. The objective is not to transform the Won into an alternative reserve currency, nor to replicate China's offshore RMB model. Rather, policymakers are focused on creating the conditions for a more internationally usable currency — one that is easier to trade, fund, settle and hold across time zones. The roadmap combines the development of offshore settlement and funding infrastructure with a gradual easing of selected capital-account restrictions, allowing international use of the Won to expand while maintaining a measured and risk-managed approach to liberalization.

More fundamentally, Korea's push to internationalize the Won reflects a broader question about how the country captures the benefits of its industrial success. Our work argued that re-industrialization, the AI infrastructure cycle and supply-chain diversification are likely to strengthen Korea's external balances and generate larger pools of national savings. Yet generating wealth and retaining wealth are not the same thing. A significant share of the savings generated by Korea's industrial success continues to be recycled abroad through portfolio investment and overseas asset accumulation. **Strong external balances therefore do not automatically translate into deeper domestic capital markets, stronger asset valuations or a more internationally important currency.**

Figure 12: The reforms started well before July 2026



Source : Deutsche Bank Research

In many respects, Won internationalization can be viewed as the financial counterpart to Korea's industrialization. Industrialization generates exports, profits, savings and external surpluses; internationalization helps deepen capital markets, broaden participation and increase the likelihood that a greater share of the resulting trade, investment and funding activity is intermediated through Korean financial markets. **The objective is therefore not simply a more international currency, but a financial system that more closely reflects Korea's growing role in the global economy.** The result is not a wholesale opening of the capital account, but a gradual and carefully sequenced effort that combines greater offshore usability of the Won with the selective liberalization of domestic financial markets, while preserving financial stability and policy flexibility.

Korea's model – A phased approach to internationalization

A common mistake is to analyze Korea's internationalization strategy primarily through the lens of either China or Singapore. In reality, Korea is pursuing a distinct path that differs from both models.

China's approach relied heavily on the creation of a parallel offshore ecosystem that allowed the renminbi to internationalize while maintaining extensive control over domestic financial markets. Singapore represents the opposite approach, combining deep global integration with virtually unrestricted capital mobility. Korea's strategy sits somewhere between these two models. Policymakers are seeking to expand the usability of the Won while preserving financial stability and policy flexibility.

The first phase of reform focused on improving market accessibility. Measures such as the Registered Foreign Institution framework, omnibus accounts, extended FX trading hours and broader market-access reforms were designed to reduce operational barriers facing global investors and improve access to Korean financial assets. These reforms laid the foundations for greater foreign participation in Korean bond and equity markets.

The next phase goes further by focusing on the usability of the currency itself. Recent reforms are designed to make the Won easier to hold, transfer, fund and settle internationally. Rather than concentrating solely on attracting investors into Korean assets, policymakers are now seeking to expand the range of activities that can be conducted in Won across investment, funding and settlement activities.

Importantly, liberalization is being implemented gradually. Korea is not pursuing a wholesale opening of the capital account. Instead, reforms are being sequenced carefully alongside monitoring systems, liquidity facilities and macroprudential safeguards. The objective is to reduce frictions and improve accessibility without compromising financial stability.

The result is best understood as a phased transition from market-access reform towards broader currency internationalization. Earlier reforms focused on making Korean financial assets easier for global investors to access. The next phase is aimed at making the Won itself easier to use across investment, funding and settlement activities.

Figure 13: Korea is pursuing a distinct model of currency internationalization

Korea is pursuing a distinct model of currency internationalisation

	Singapore	China	India	Korea
Primary objective	Global financial centre	Global reserve currency	Expand INR use in trade and finance	Increase KRW investability and global usage
Internationalisation model	Full openness	Offshore-onshore framework (CNH/CNY)	Trade-led internationalisation	Phased internationalisation led by capital-market integration
Key policy tools	Open capital account, global financial intermediation	Offshore RMB markets, clearing banks, Connect programmes	INR trade settlement, bilateral arrangements	FX market liberalisation, benchmark inclusion, settlement infrastructure
Treatment of offshore markets	Integrated	Separate offshore ecosystem	Limited	Greater offshore usability through infrastructure linked to the domestic market
End-state vision	Fully open financial centre	Widely used international currency	Broader regional and trade usage	Globally investable and increasingly usable currency
Key challenge	Small domestic market	Capital controls	Limited global demand	Balancing openness and financial stability
Sequencing	Openness first	Trade settlement → Offshore ecosystem → Investment integration	Trade settlement → Gradual financial liberalisation	Market access → Broader currency usage

Source : Deutsche Bank Research

How Korea is building a more international Won?

The first pillar is market accessibility. Korea has already moved to 24-hour dollar-Won spot trading, allowing investors to trade and hedge Won exposure across Asian, European and North American trading sessions. Earlier reforms such as the Registered Foreign Institution (RFI) framework, omnibus accounts and settlement-system upgrades were designed to reduce many of the operational frictions that historically limited foreign participation in Korean financial markets. These measures laid the foundations for greater foreign ownership of Korean bonds, equities and other financial assets.

The second pillar is offshore usability. The proposed framework allows registered Offshore Won Settlement Institutions to facilitate Won transfers, funding and settlement between non-residents without requiring every transaction to pass through traditional domestic account structures. A new BoK offshore settlement network is intended to support 24-hour settlement and improve the ability of investors, corporates and financial institutions to hold, transfer and deploy Won outside Korean business hours. Authorities are also seeking to encourage greater use of deliverable KRW markets through offshore settlement infrastructure and measures that support deliverable rather than non-deliverable instruments over time. This reflects one of the central features of the roadmap: expanding the ability of non-residents to hold, transfer, fund and settle Won offshore, even as onshore liberalisation remains gradual and carefully sequenced.

The third pillar is gradual liberalization. Rather than removing all restrictions at once, policymakers are selectively easing reporting requirements, simplifying account structures and gradually shifting certain transactions from prior approval to ex-post reporting. Importantly, liberalization is occurring within a controlled framework: offshore Won transactions initially operate through registered institutions, while market monitoring, liquidity facilities and prudential safeguards are expanded in parallel.

Figure 14: The building blocks of KRW internationalization

Pillar	What has changed	Why it matters
Market accessibility	24-hour USD/KRW trading; RFI framework; omnibus accounts; settlement upgrades	Allows foreign investors to access, trade and hedge Korean assets more easily across global time zones
Capital market integration	WGBI inclusion, MSCI market-access reforms, Euroclear/ICSD connectivity, expanded collateral and repo infrastructure	Reduces operational barriers for global investors and integrates Korean assets more closely with global portfolios
Offshore won usability	Offshore Won Settlement Institutions and offshore settlement network	Allows non-residents to hold, transfer, fund and settle won outside Korea more efficiently
Regulatory liberalisation	Many inter-foreigner won capital transactions move toward lighter reporting and ex-post supervision	Reduces frictions for investors, corporates and treasury centres while maintaining oversight
Digital money infrastructure	Stablecoin framework, CBDC initiatives, tokenised government-bond pilots	Establishes future rails for cross-border payments and digital won usage

Source : Deutsche Bank Research

Taken together, these reforms represent an evolution from market-access reform towards broader currency internationalization. Earlier initiatives such as the RFI framework, omnibus accounts, extended trading hours and benchmark-related reforms focused primarily on reducing barriers for foreign investors. The new roadmap builds on that foundation by expanding offshore settlement infrastructure, easing selected capital-account restrictions and increasing the range of activities that can be conducted in Won. The objective is therefore not only to make Korean assets easier to access, but also to make the Won itself easier to hold, transfer, fund and settle internationally.

Market Implication: A new funding, investment and settlement currency for Asia?

For the currency, success should not be measured by whether the Won becomes a major reserve currency. **A more realistic benchmark is whether the Won gradually plays a larger role in economic activities linked to Korea's own industrial and trading footprint.** Progress would likely be visible through greater use of Won settlement in trade, broader participation by foreign investors in Korean financial markets, and increasing use of Won-denominated funding and hedging instruments. The reforms create the infrastructure necessary for such a transition, but market adoption will ultimately determine the outcome.

Figure 15: Measuring progress towards a more international Won

Success Metric	Korea	Japan	Australia	Why it matters
Foreign ownership of local-currency government bonds	28.7%	13.7%	52.6%	Higher foreign participation and broader investor base
Foreign ownership of domestic equities	38.5%	11.9%	30.9%	Increased representation in global portfolios
Institutional ownership of equities	20.0%	34.0%	60.0%	More stable portfolio flows
Inclusion in Global DM benchmark	Not in MSCI DM	Yes	Yes	Measures potential structural demand from its financial assets
Home-currency settlement share in external trade	3.0%	33% (exports only)	8.8%	Gradual increase in trade invoicing and settlement in won
Foreign participation in repo / funding markets	2.1%	N/A	60.0%	Deeper offshore funding and collateral markets
Currency share of global FX turnover	0.9%	8.5%	3.1%	Greater turnover and broader participation

Source : Deutsche Bank Research

The most immediate market impact may be a gradual migration from NDFs toward deliverable KRW markets. Improved access and offshore settlement infrastructure should support greater use of deliverable KRW instruments over time, although NDFs are likely to remain an important part of the ecosystem. The goal is not to eliminate NDFs, but to create more attractive deliverable alternatives, deepen liquidity and strengthen the link between offshore activity and Korean financial markets.

The implications may extend beyond the currency itself. A more internationally accessible financial system could support deeper capital markets, stronger liquidity and a broader investor base for Korean assets. Over time, this may help narrow valuation discounts that persist despite Korea's strong industrial position, improving the linkage between the country's economic performance and financial-market performance. In this sense, internationalization should be viewed not merely as an FX-market reform, but as part of a broader effort to align Korea's financial influence more closely with its growing economic and industrial importance.

Ultimately, Korea's post-war success was built on industrialization. Re-industrialization, AI investment and supply-chain diversification suggest that story is far from over. The next decade may be defined by a different question: whether Korea can convert industrial power into financial power. The success of internationalization will therefore be judged not by reserve-currency status, but by whether Korea narrows the gap between its economic footprint and its financial footprint. If industrialization was the first stage of Korea's development story, internationalization may prove to be the second.

Appendix 1

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The Expanding Frontier of Agentic AI: A User's View

August 6, 2026

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The most important change in AI today may not be the arrival of a single new frontier model, but the speed at which frontier-like capability is becoming cheaper. My own use of AI agents since March suggests the shift is already visible: models that once looked like expensive high-end choices are being matched by cheaper alternatives within months. This report looks at that expanding frontier and what it means for adoption.

Since I started using AI agents in March, trying new models has become something of a hobby—and partly a necessity. I could not rely on the most advanced models to power my OpenClaw Bots and other tools, not only because access is not always guaranteed, but also because they are extremely expensive for agentic usage. I therefore spent the past few months switching among more accessible and affordable alternatives, many of them models developed by Chinese labs.

The effect of each switch was often not obvious. A model might seem more reliable or require less instruction, but the improvement was difficult to separate from chance. Over a longer period, however, the change became unmistakable. My AI agents could complete the same tasks with fewer prompts, use tools more effectively and travel further from a simple instruction before requiring intervention.

That said, two recent model releases did feel different to me. Both Kimi-K3 and DeepSeek-V4-Flash-0731 seemed notably more powerful than the previous alternatives I had been using. Was this merely a subjective experience, or had the underlying models genuinely improved? This prompted me to do some databased analytical research.

Measuring agentic intelligence against cost

The research I did was a simple plot of popular models along two dimensions: capability and cost. The horizontal axis is API cost per million tokens on a log scale, using a simple blend of 90% input tokens and 10% output tokens. The vertical axis measures model capability. There are multiple possible measures, but I chose the Agent Index from LLM Stats for two reasons: first, I care more about agentic capability than other dimensions such as coding or writing because of my use case; second, LLM Stats offers a composite measure based on benchmarks involving tool use, multi-step execution, web search, terminal operation, software engineering and other agentic tasks.

The broad pattern is intuitive: more capable models tend to be more expensive. US models are shown in blue, and Chinese models in red in the chart. A clear upward correlation can be observed across all the dots. The difference is that US models occupy much of the upper-right part of the chart, while Chinese models are more densely represented at lower price points. The regression lines capture this difference: both groups exhibit a positive

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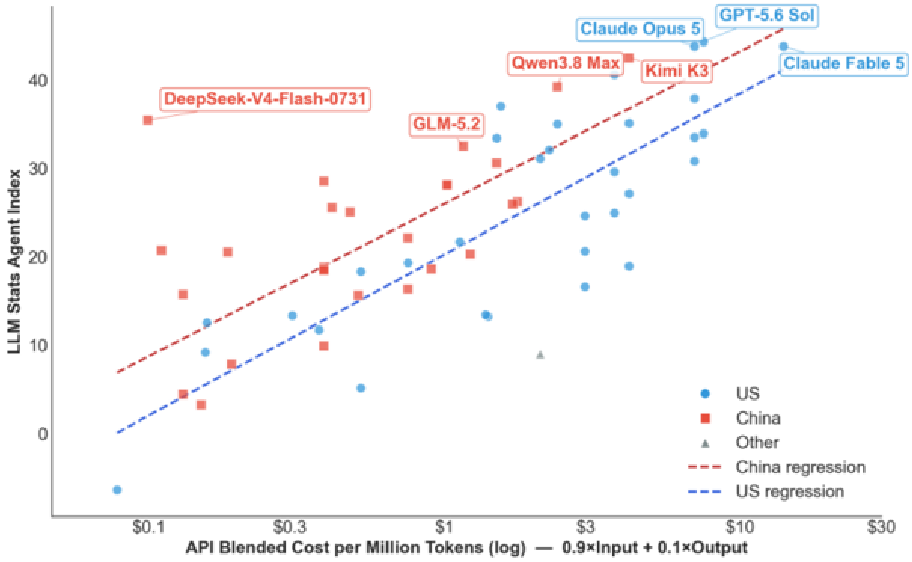
relationship between cost and agentic performance, but the US cluster generally extends further into the highcost, high-capability region.

US and Chinese AI Models copared on cost and Agentic capability



Source: Deutsche Bank Research, LLM Stats

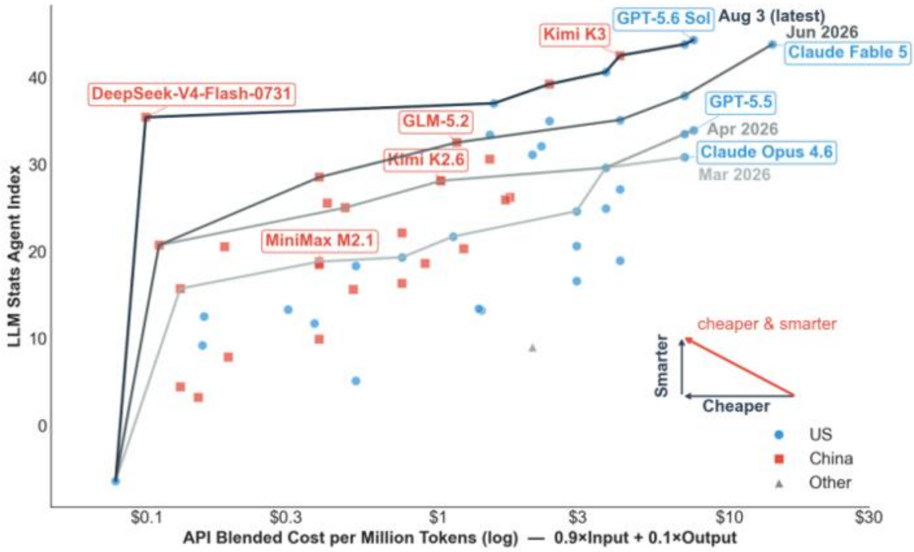
A frontier moving rapidly

The more useful concept for model selection—at least for me as an economist— is the Pareto frontier. A model is Pareto optimal if no cheaper model has a higher score. Connecting all Pareto optimal models gives the Pareto frontier: the boundary of best available model performance at each cost level. A model below that frontier may still suit a particular task, but on these two dimensions it is dominated by another choice. The chart below shows the Pareto frontier for all models in my dataset.

For example, at the top of the current frontier, GPT-5.6 Sol has the highest Agent Index in the dataset, at 44.3, with a blended cost of $7.50 per million tokens (the dot next to it is Opus 5). This is just too expensive for me. If I want to reduce cost, the next best choice according to the chart would be Kimi-K3: with only a modest loss in capability, it offers a meaningful cost saving and could be a good choice for slightly less complex tasks. But it is still too expensive for simple tasks, for which I would go even further down the line to maybe DeepSeek-V4-Flash.

But the analysis did not stop there. The chart also shows how the Pareto frontier has shifted as new models became available. By including only models released by each cutoff date — March 31, April 30, June 30 and August 3 — I was able to reconstruct earlier frontiers and compare them with the latest one.

Expanding Pareto Frontier: Agentic AI is becoming cheaper and smarter at speed



Source: Deutsche Bank Research, LLM Stats

The speed of Pareto improvement is striking. Every month, the Pareto frontier moves upward and left by a large margin. This means users can achieve better performance at the same cost, or cut spending significantly while maintaining similar performance. For example, GLM 5.2 on the June frontier has almost the same score (32.5) as Opus 4.7 (33.5) on the April frontier, while its blended cost of $1.16 is far below Opus 4.7's $7. In merely two months, a user could potentially reduce token cost by more than 80% without losing performance.

The latest frontier line makes the point even more clearly. As of today, almost all models on the latest Pareto frontier have exceeded GPT-5.5's score. In other words, **what sat on the high-cost frontier in April is now available across the cost spectrum.**

Chinese AI models are catching up

It would be easy to summarize the chart by saying that US models remain ahead… At the absolute frontier, that is still true: GPT-5.6 Sol holds the highest Agent Index in the dataset. But that reading misses the more consequential competitive dynamic.

…but Chinese models' advancement from the lower-left is also remarkable. In March, when I first started using AI agents, the Pareto frontier was dominated by US models. Kimi K2.6 moved into the middle by April; GLM-5.2 approached the performance of recent leaders by June; and Kimi K3 narrowed the gap near the top in July. Kimi K3, being the largest (2.8T parameter) and most costly openweight model to date, shows that Chinese models are no longer competing only on price; they are increasingly present near the capability frontier as well

The DeepSeek Outlier

What strikes me most is how the latest DeepSeek release altered the shape of the chart. DeepSeek-V4-Flash-0731, released last week, was a large improvement over the earlier low-cost version included in the April frontier. The wearlier model was already inexpensive, with an Agent Index of 20.7 at $0.11 per million tokens. The July release has effectively the same cost, but its Agent Index rises sharply to 35.4. It moves DeepSeek from the low-cost edge of the chart into the performance range of much more expensive frontier models.

DeepSeek demonstrates how much potential remains for AI models to improve. DeepSeek-V4-Flash is far smaller than models near the current frontier. At 284B parameters in total (13B active under a Mixture-of-Experts architecture), its agentic capabilities outperformed much larger models such as the 744B GLM-5.2 and the 1.6T DeepSeek-V4-Pro. The improvement was achieved not through increasing model size (making it more knowledgeable) but through post-training (teaching it to better handle tasks). The contrast illustrates that good agent performance does not require the largest model; architecture and post-training can move the effective frontier more than parameter count alone would suggest.

The Jevons Paradox is Working on Me

Another interesting fact: the steep decline in AI model costs did not bring me any savings. The opposite is happening—my AI bills have surged. My total spending on AI was effectively zero in February before I started using AI agents. Then I started using OpenClaw and made my first token subscription plan at $20/month. A few months later, my recurring spending on AI is now reaching $100-200/month. My costs increased because I now use AI agents to do many more things and must subscribe to new plans as tokens run out under existing ones. The increase also includes cloud storage and an $800 Mac Mini to host my "little lobsters" (OpenClaw Bots).

This is a familiar economic mechanism: the Jevons paradox. Improvements in efficiency did not reduce consumption; rather, they encouraged wider use and led to an increase in total usage and total spending. As model capability improved while token cost dropped, I expanded the use of my AI agents to more daily tasks—tracking headline news and specific topics I am interested in, summarizing market movements, researching academic papers, managing expenses, making travel plans and testing ad-hoc ideas. Why use a chatbot if you have dedicated AI agents that know your background and remember the conversation history?

The question is therefore not simply which model will hold the highest score next month. It is what happens when millions of users and businesses can afford to keep capable agents running for longer, on more tasks, with less concern about every token they consume. If the cost of agentic intelligence continues to fall while its capability rises, will efficiency save tokens—or will Jevons' paradox help trigger an explosion in real-world AI use?

My verdict: the expansion of the AI Pareto frontier is benefiting users and the broader AI ecosystem. As US and Chinese labs compete, premium frontier models push the performance ceiling higher, while efficient lower-cost models make agentic workflows economical at scale. Together, they allow users to match each model to the value and difficulty of the task, rather than accept a single price-performance trade-off. Competition is accelerating the race, while the global circulation of research ideas, open weights and practical techniques is helping improvements spread quickly across the field. **The analysis made me smore confident that the future of agentic AI is arriving faster than many expected.**

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